

# Cross-Wearer Gesture Recognition — LOUO Variant Comparison

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## Data & configuration

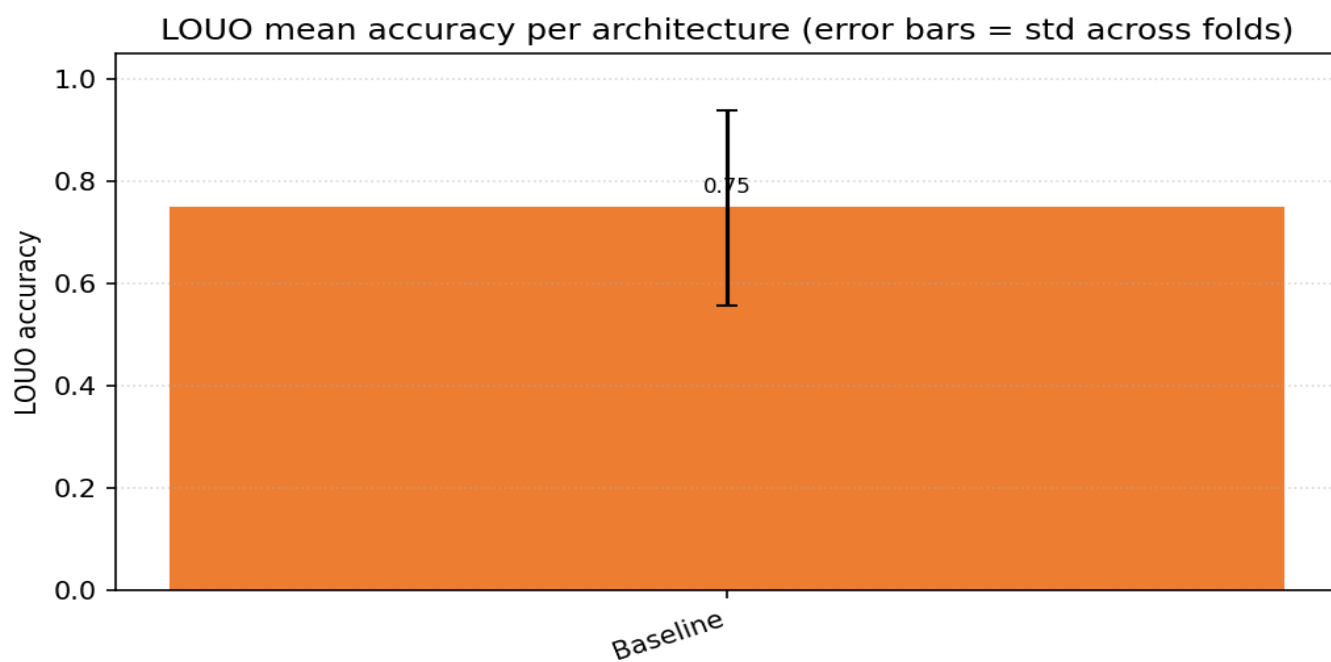
|                             |                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| LOUO path 1                 | /home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic                                                                                     |
| LOUO path 2                 | /home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic                                                                                     |
| LOUO path 3                 | /home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic                                                                                   |
| LOUO path 4                 | /home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic                                                                                   |
| LOUO path 5                 | /home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic                                                                                    |
| LOUO path 6                 | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic                                                                                   |
| LOUO path 7                 | /home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic                                                                                   |
| LOUO path 8                 | /home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic                                                                                  |
| LOUO path 9                 | /home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic                                                                               |
| LOUO path 10                | /home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic                                                                                    |
| LOUO path 11                | /home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic                                                                                   |
| LOUO path 12                | /home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic                                                                                  |
| LOUO path 13                | /home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic                                                                               |
| LOUO users loaded           | 13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin) |
| Excluded classes            | Double_Nothing                                                                                                                            |
| Classes                     | 6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll                                               |
| Sensor channels             | 176 (sequence length 90)                                                                                                                  |
| Resample / filter           | 90 steps · Butterworth cutoff 10.0 Hz, order 4                                                                                            |
| Per-trial normalisation     | none                                                                                                                                      |
| Downstream normalisation    | minmax                                                                                                                                    |
| Sampling rate               | 30.0 Hz                                                                                                                                   |
| Random state                | 42                                                                                                                                        |
| Augmentation                | off                                                                                                                                       |
| Epochs · batch size         | 40 · 16                                                                                                                                   |
| Early stopping              | on · monitor val_loss · patience=8                                                                                                        |
| LOUO inner-validation       | user                                                                                                                                      |
| LOUO variants               | Baseline                                                                                                                                  |
| Final-retrain variant       | BN_GAP_Wide                                                                                                                               |
| Final-retrain held-out user | — (trained on all users)                                                                                                                  |

## Variant comparison (LOUO)

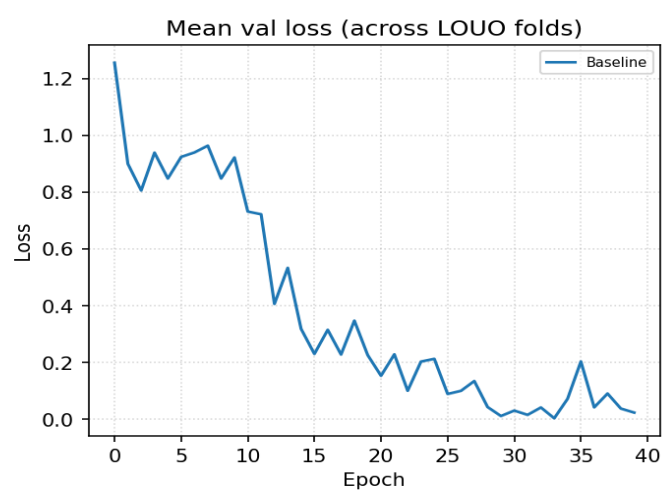
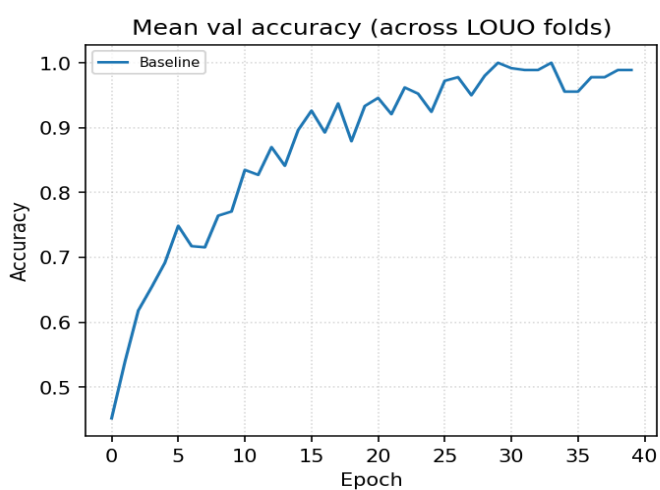
| Variant  | Params  | LOUO mean | LOUO std | Folds | Held-out user acc |
|----------|---------|-----------|----------|-------|-------------------|
| Baseline | 113,190 | 0.7494    | 0.1911   | 13    | —                 |

Best LOUO mean: <b>Baseline</b>

## LOUO accuracy per variant

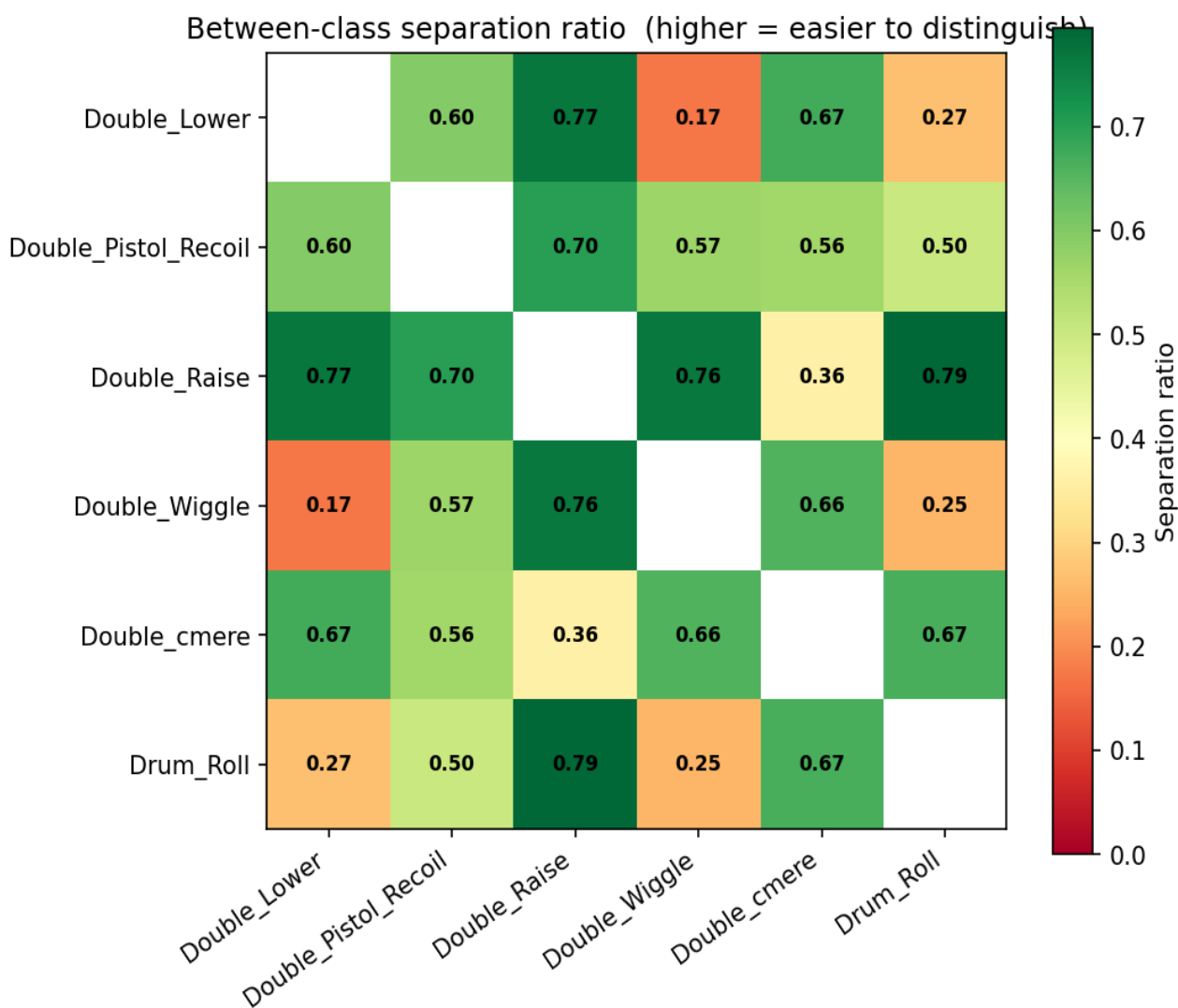
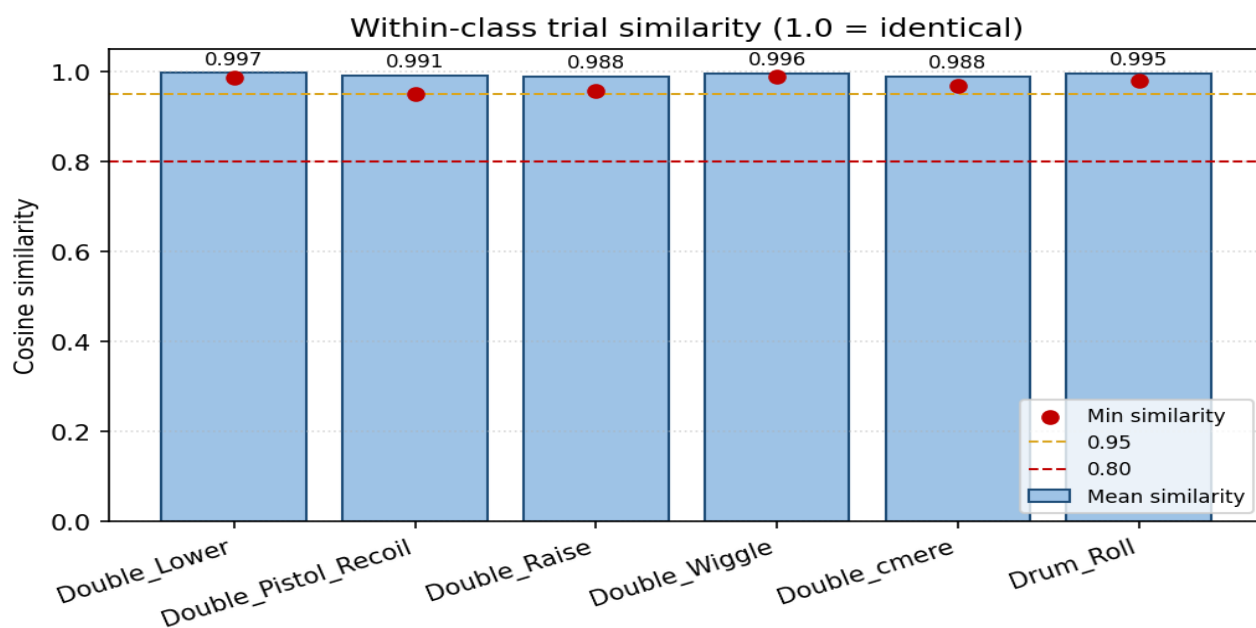


## Mean training curves across LOUO folds

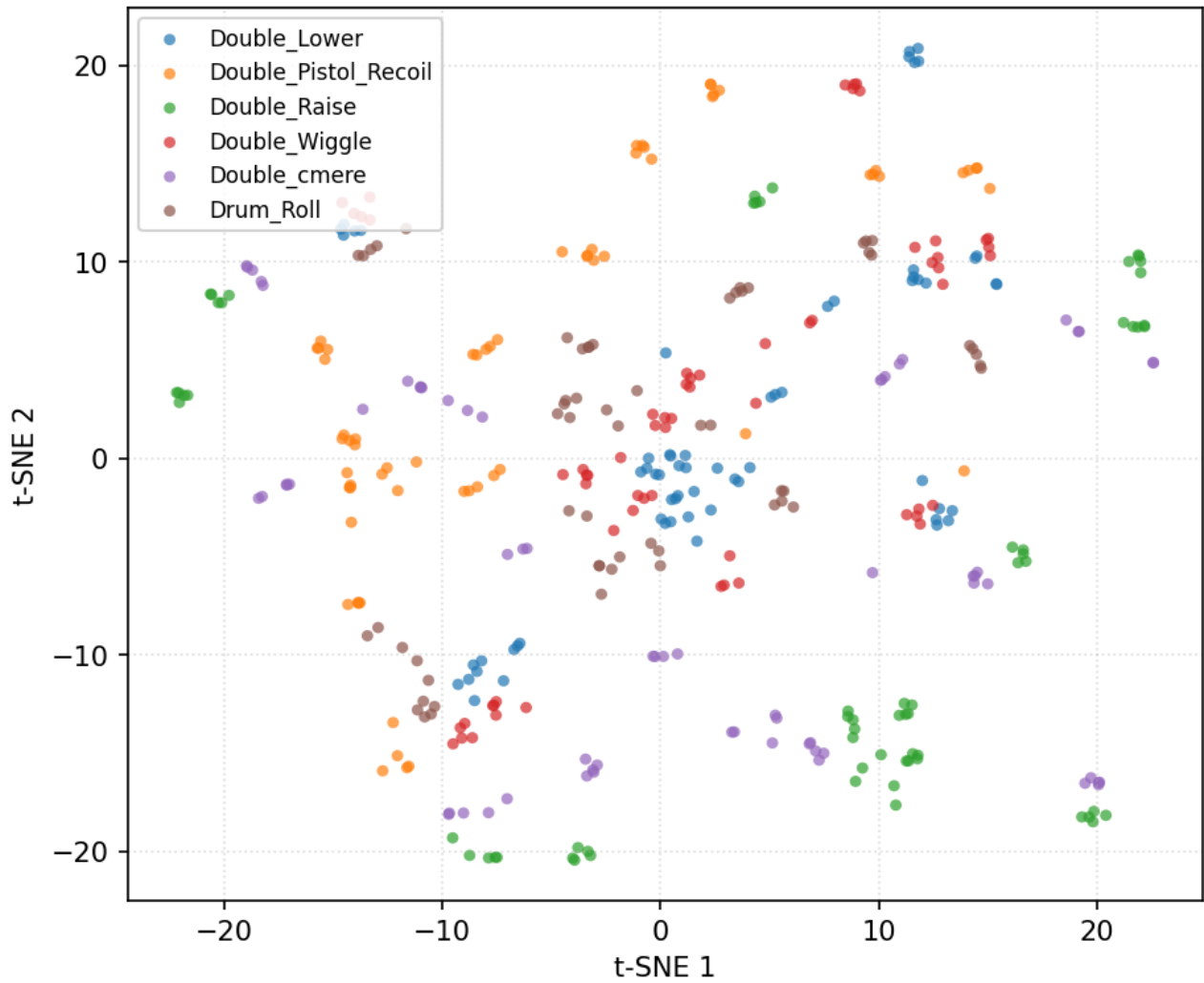


# Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).



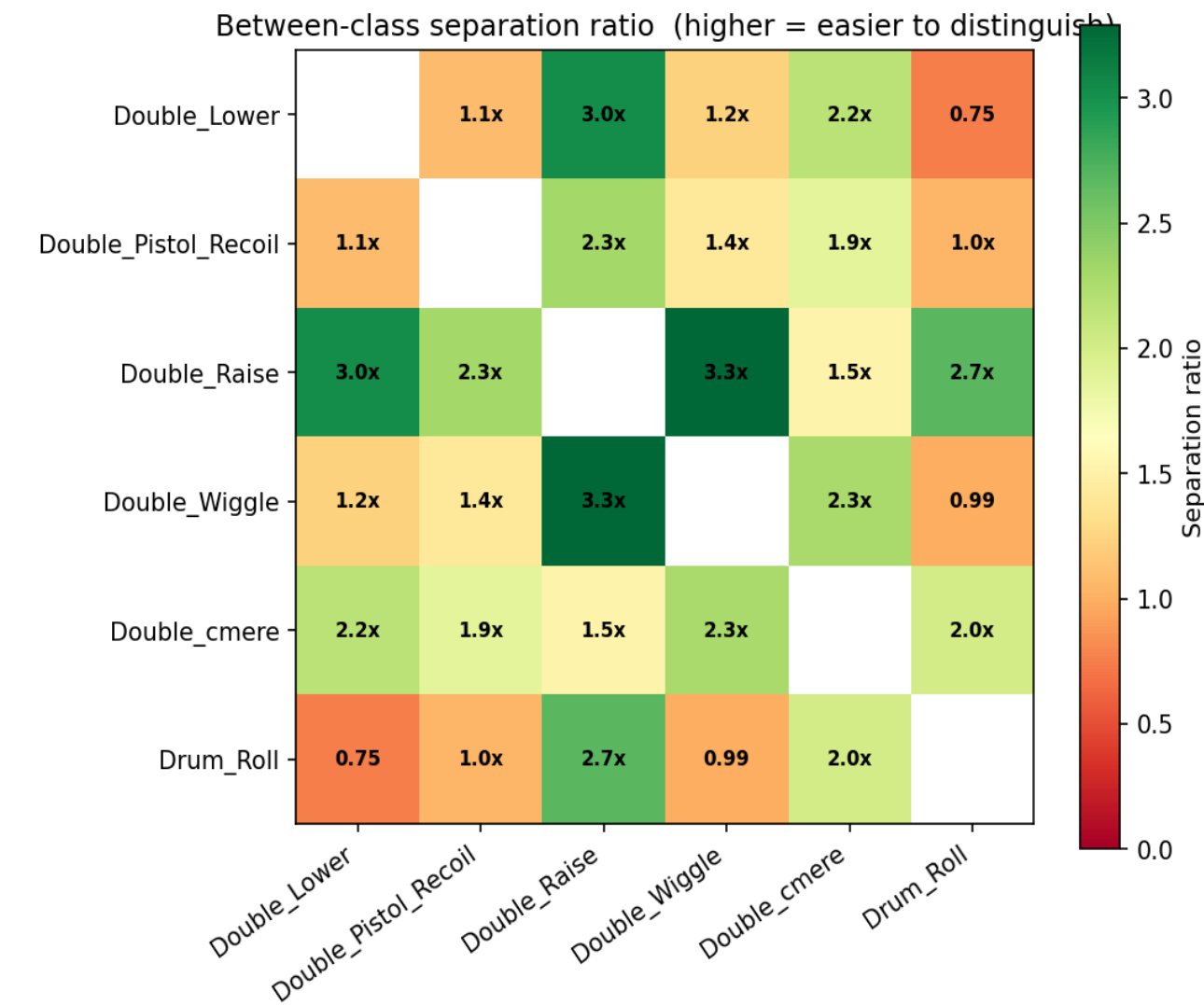
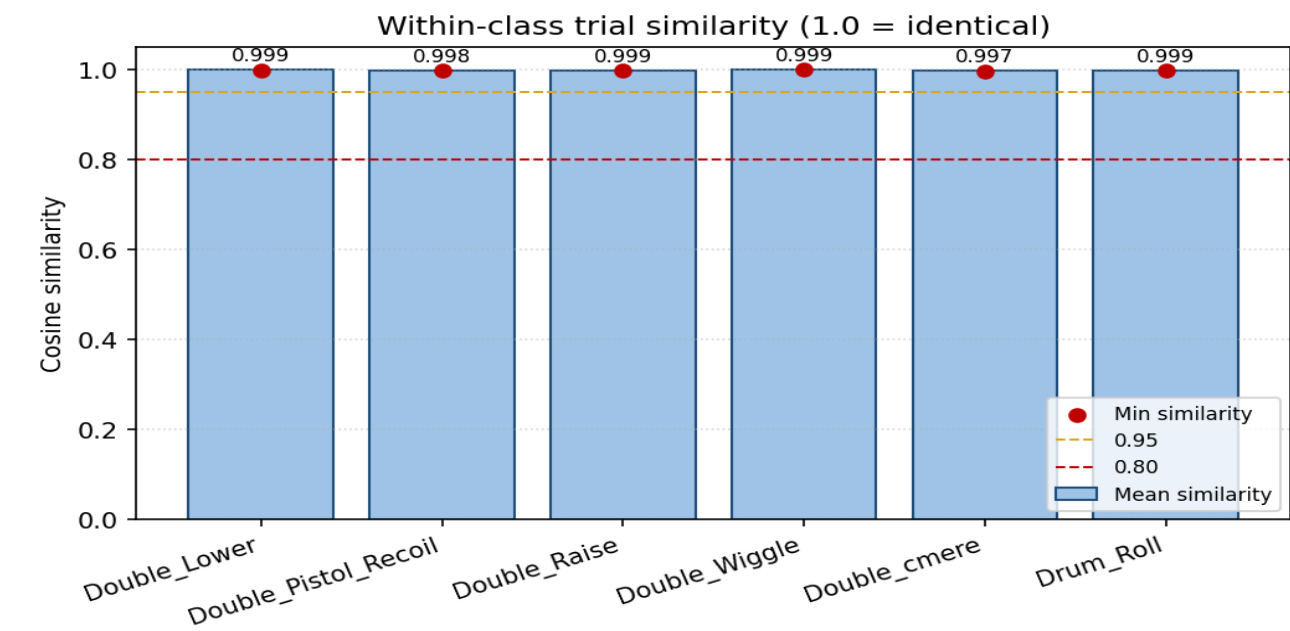
t-SNE feature space projection



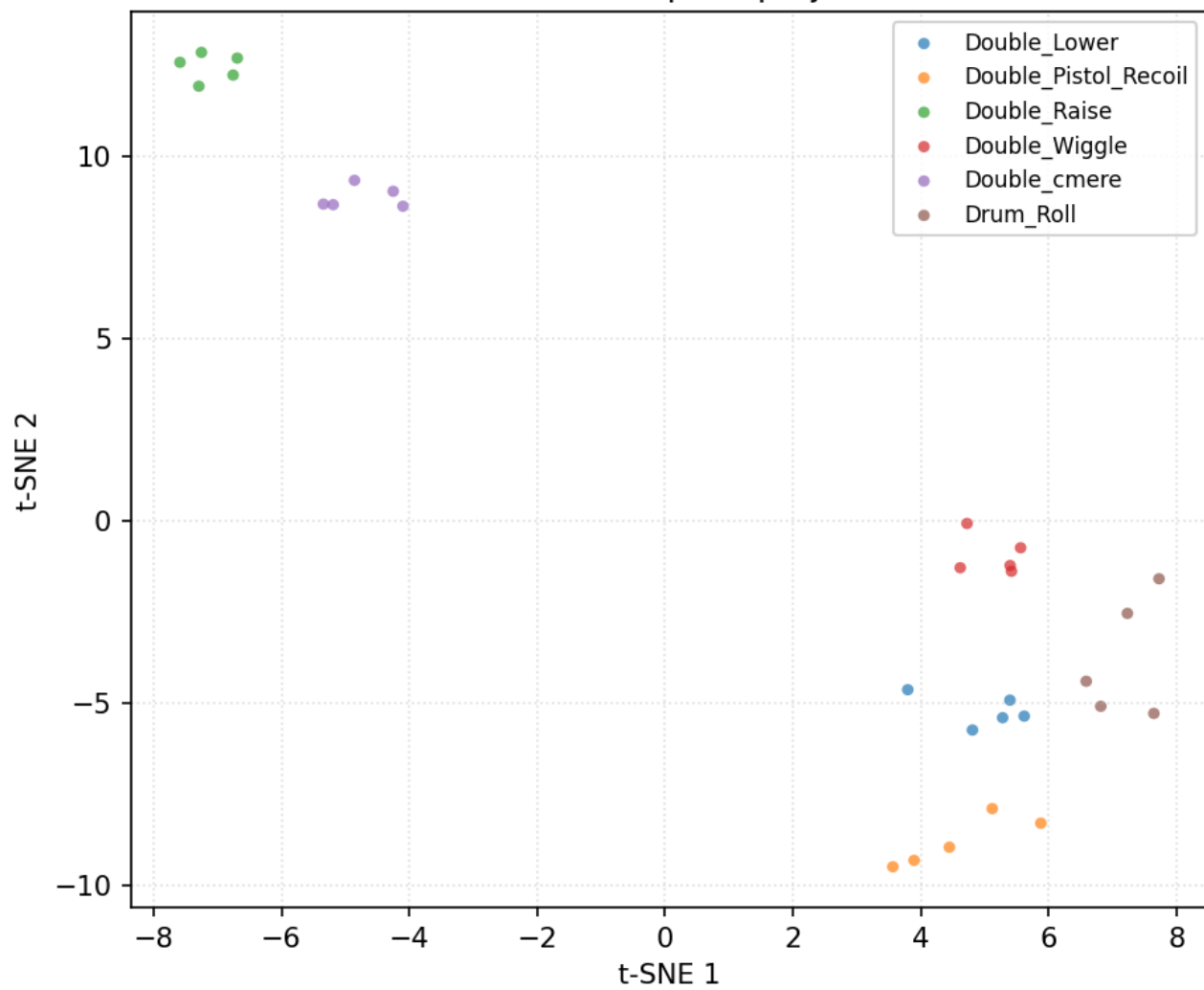
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

# Data diagnostics — user 11\_Mansh

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

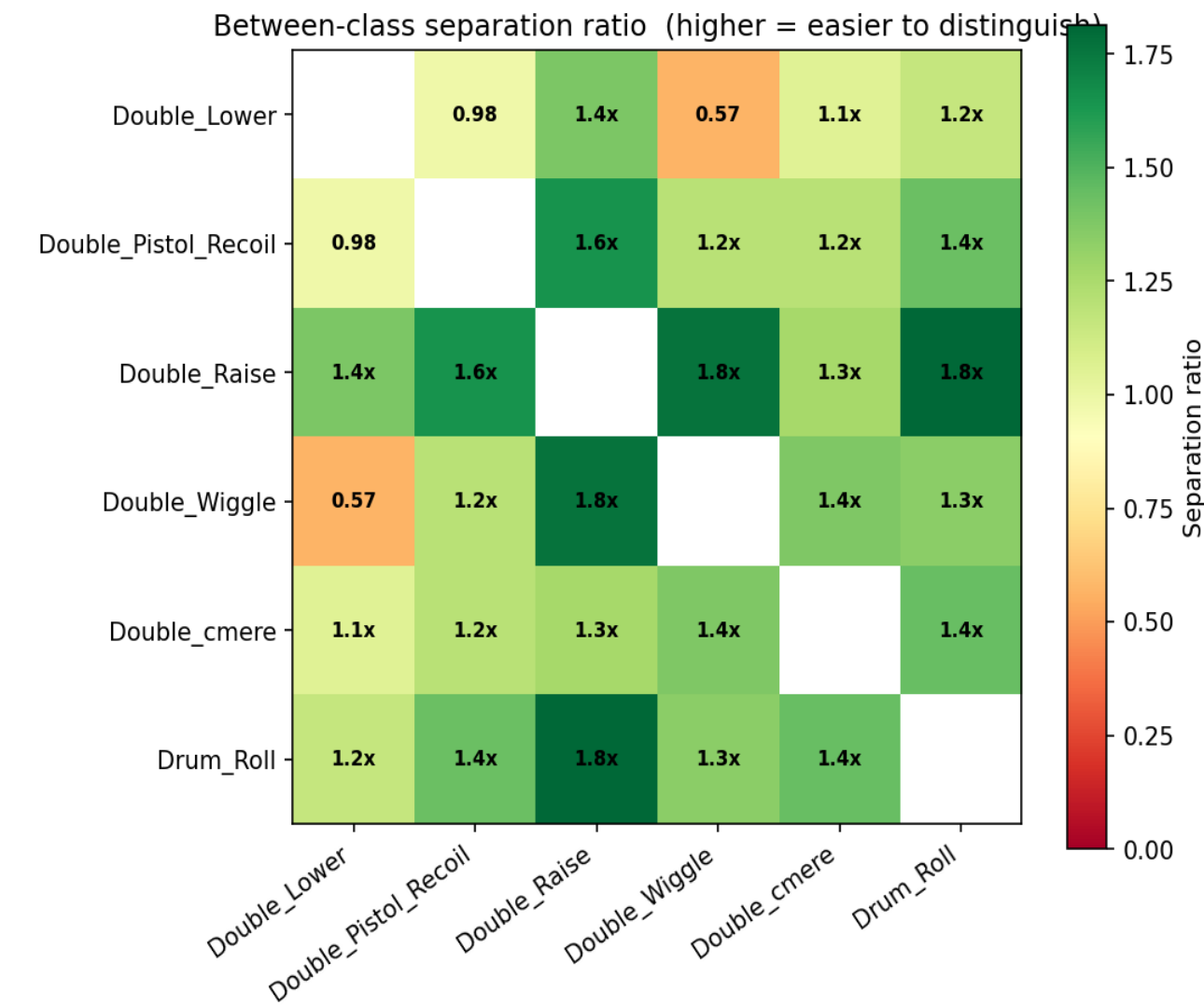
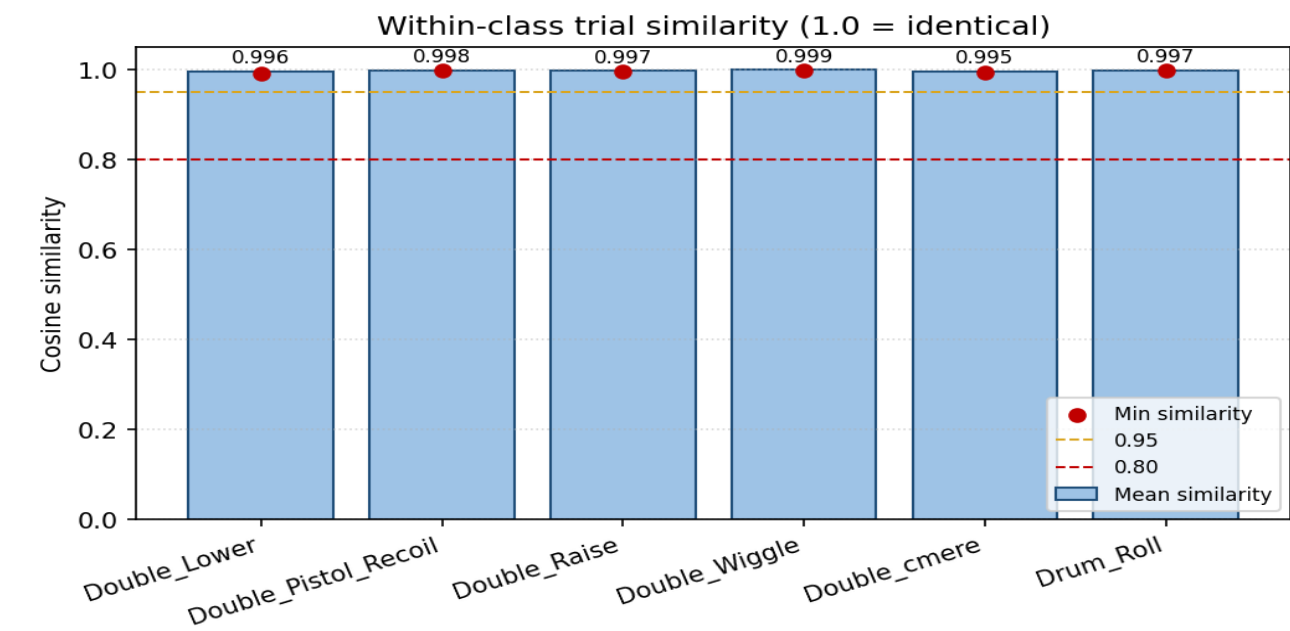


t-SNE feature space projection

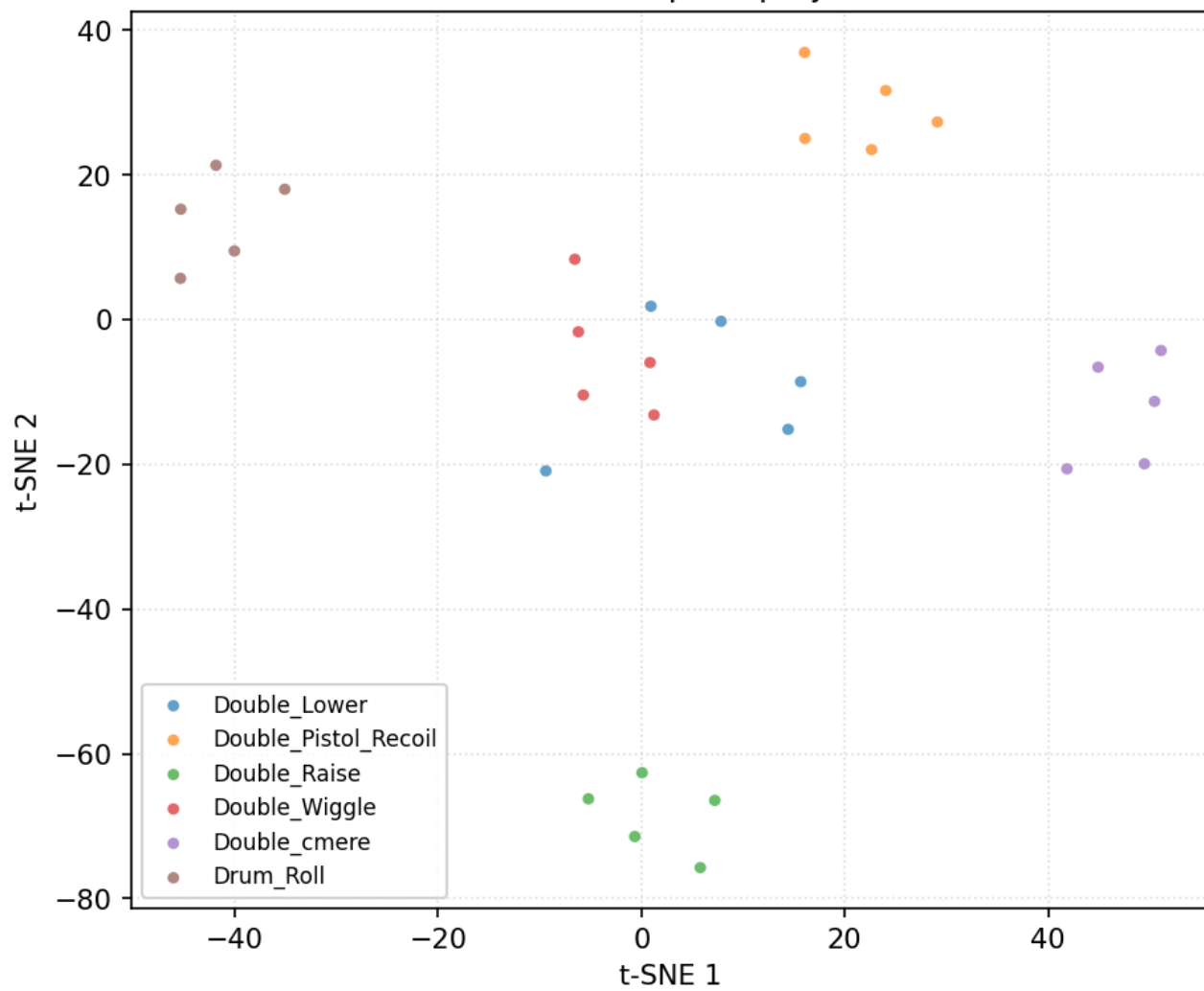


# Data diagnostics — user 12\_Marcus

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).



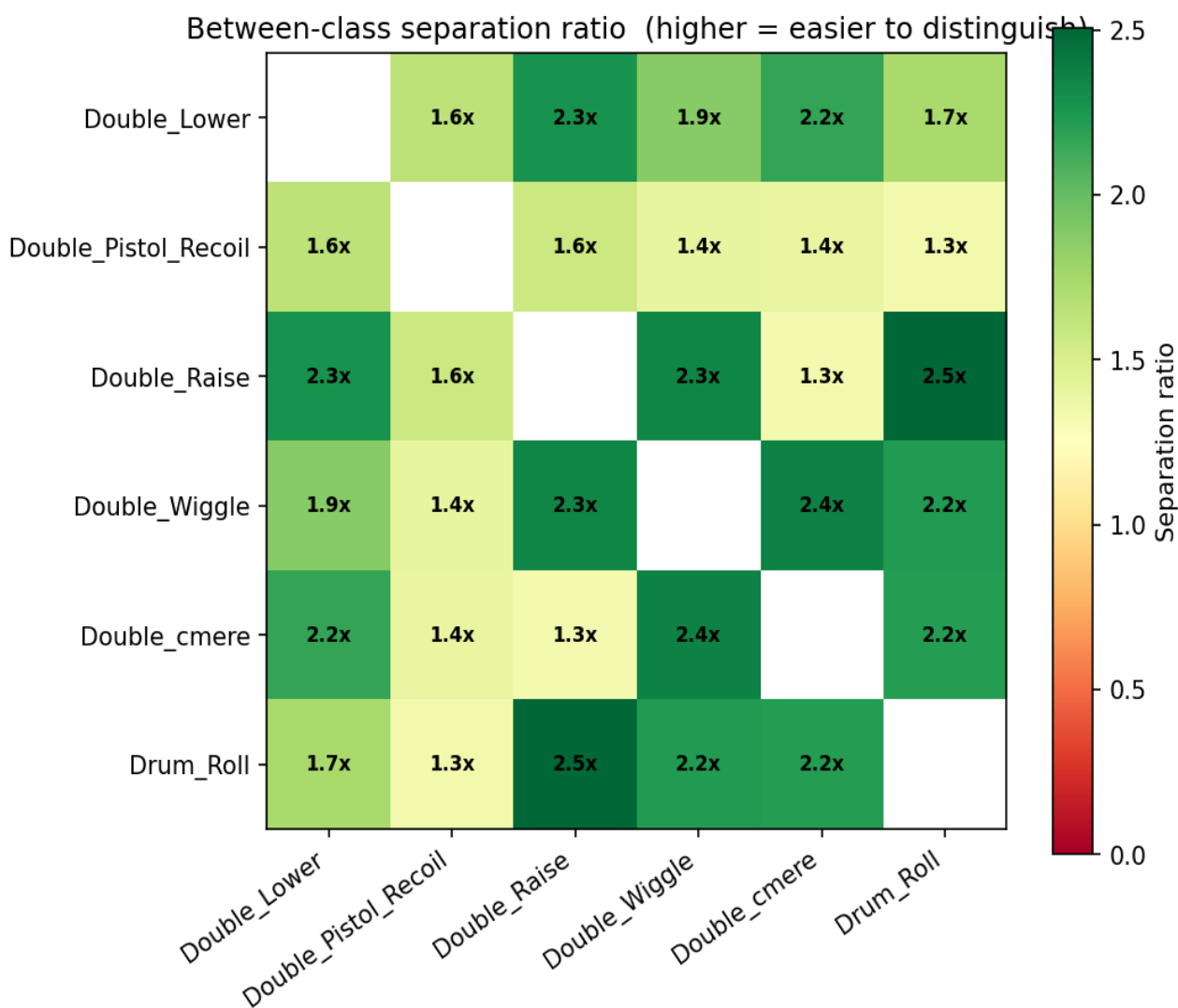
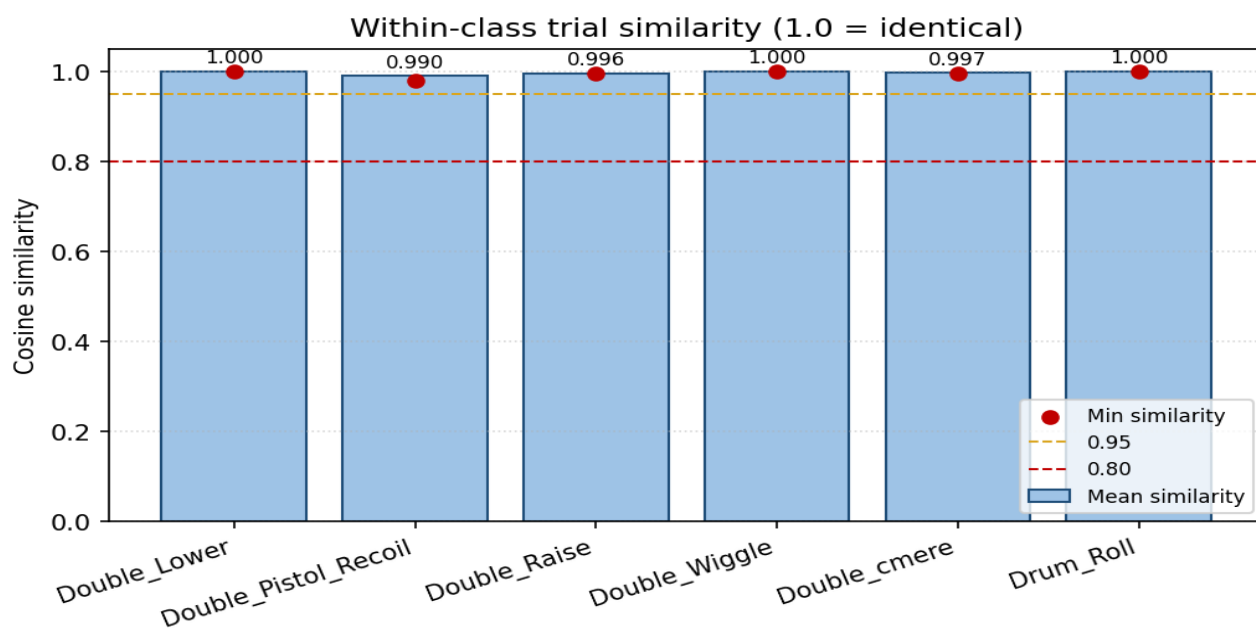
t-SNE feature space projection



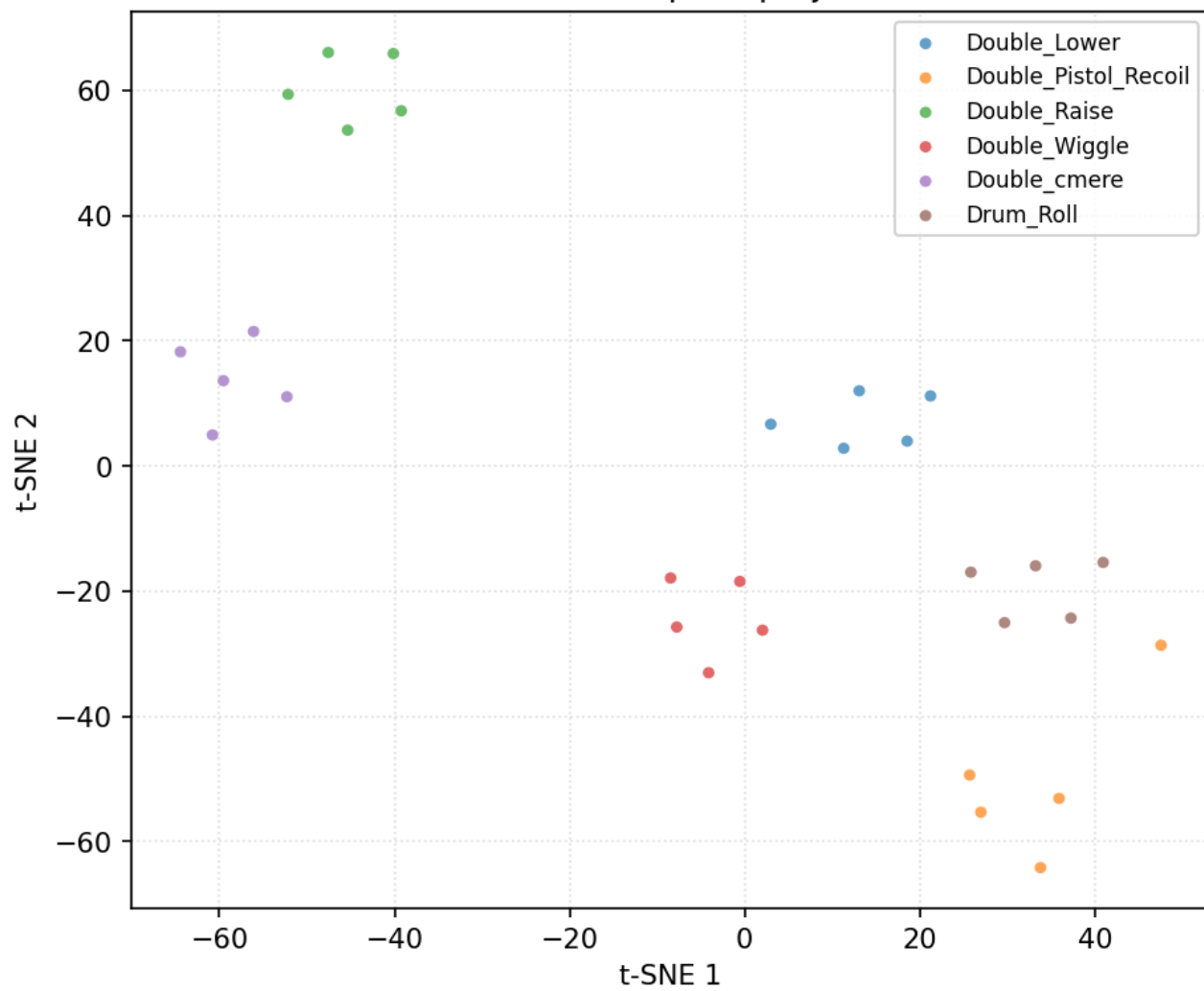


# Data diagnostics — user 14\_StephenV2

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

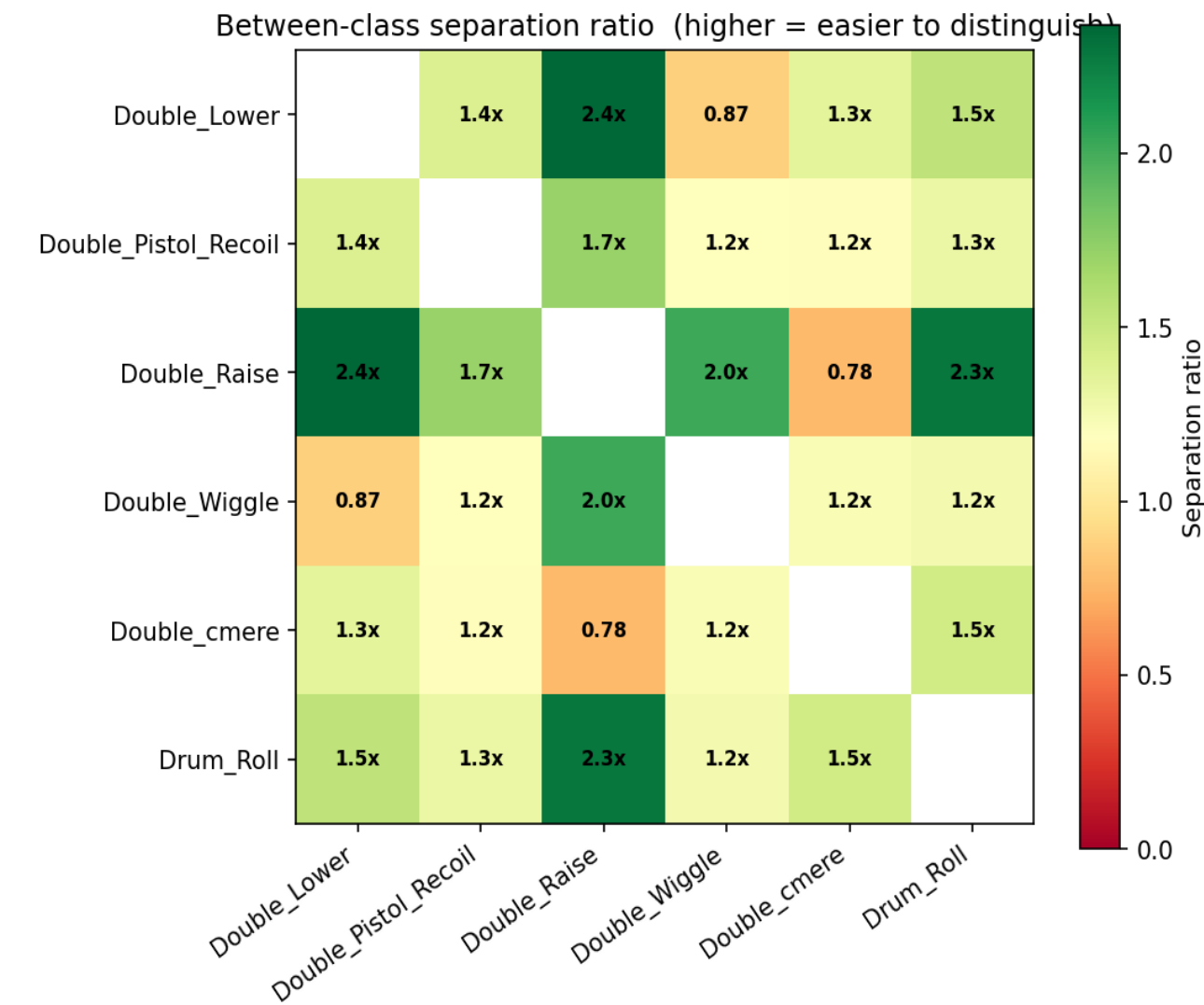
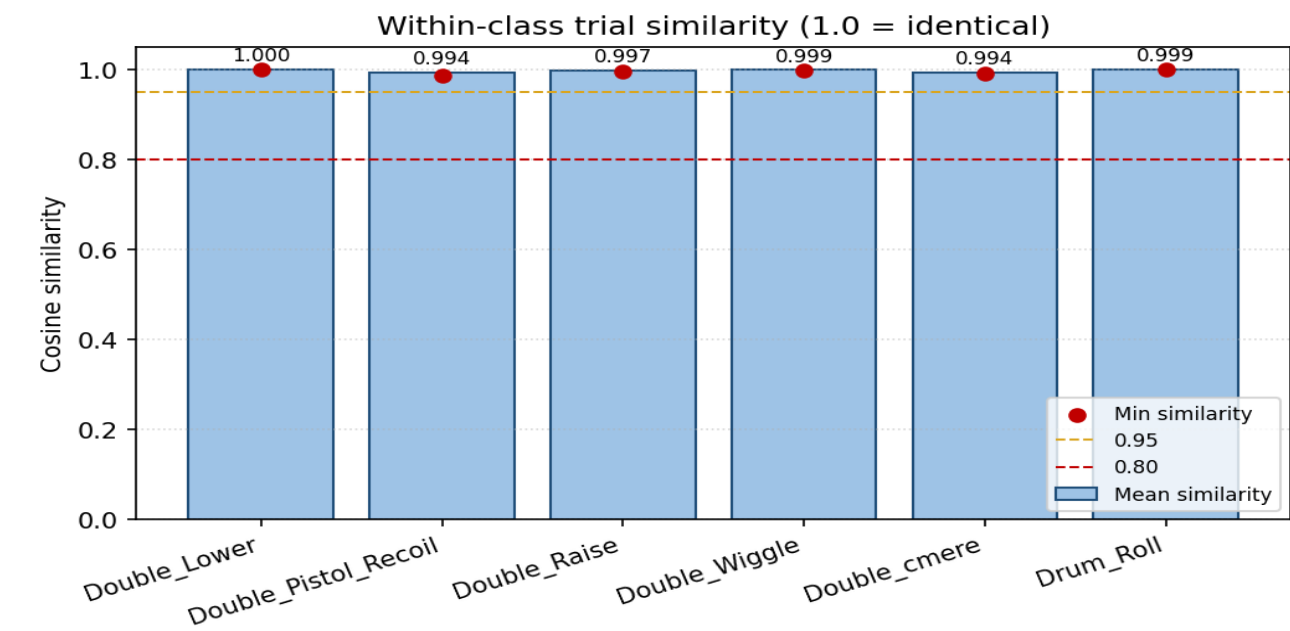


t-SNE feature space projection

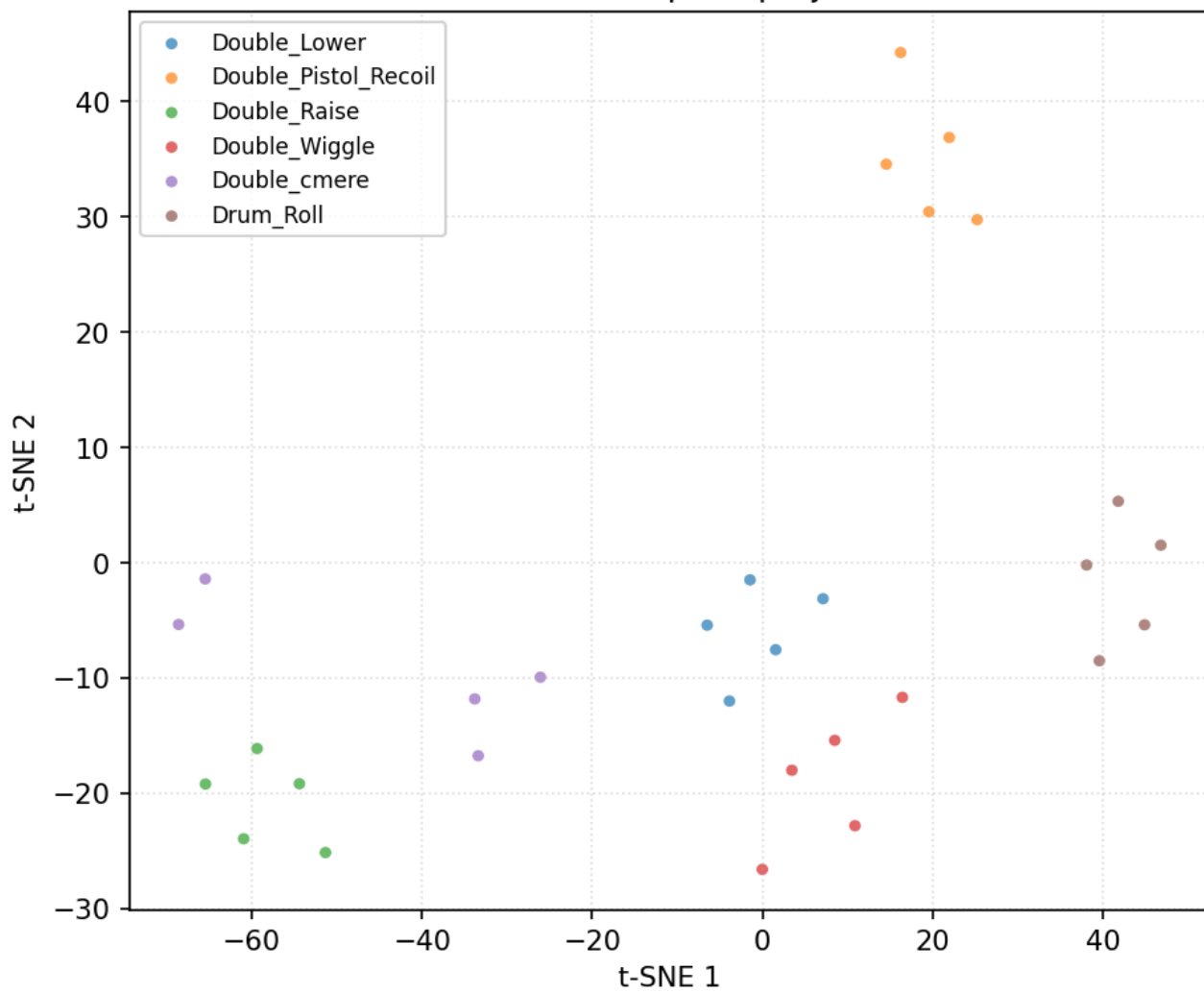


# Data diagnostics — user 15\_Tash

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

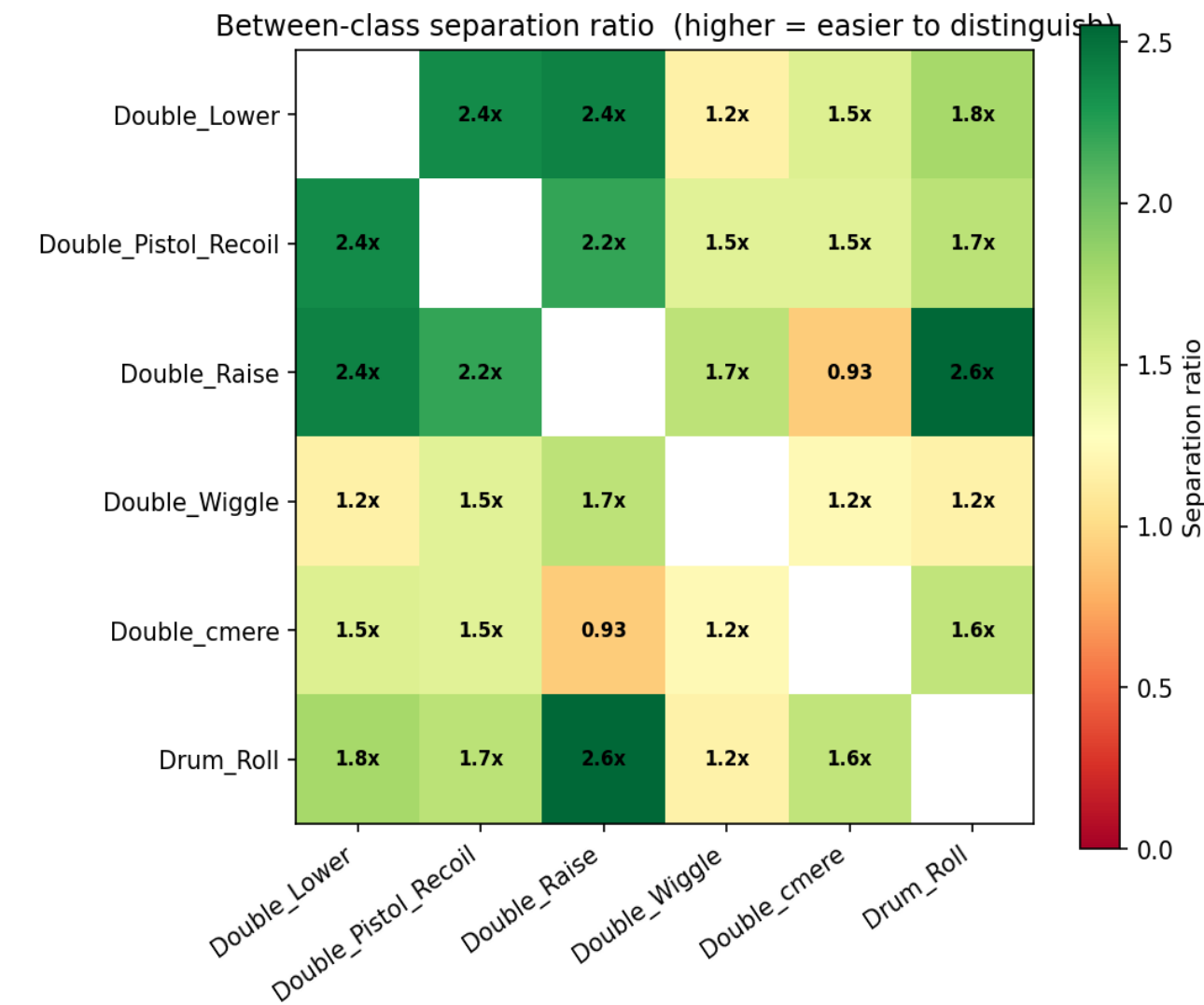
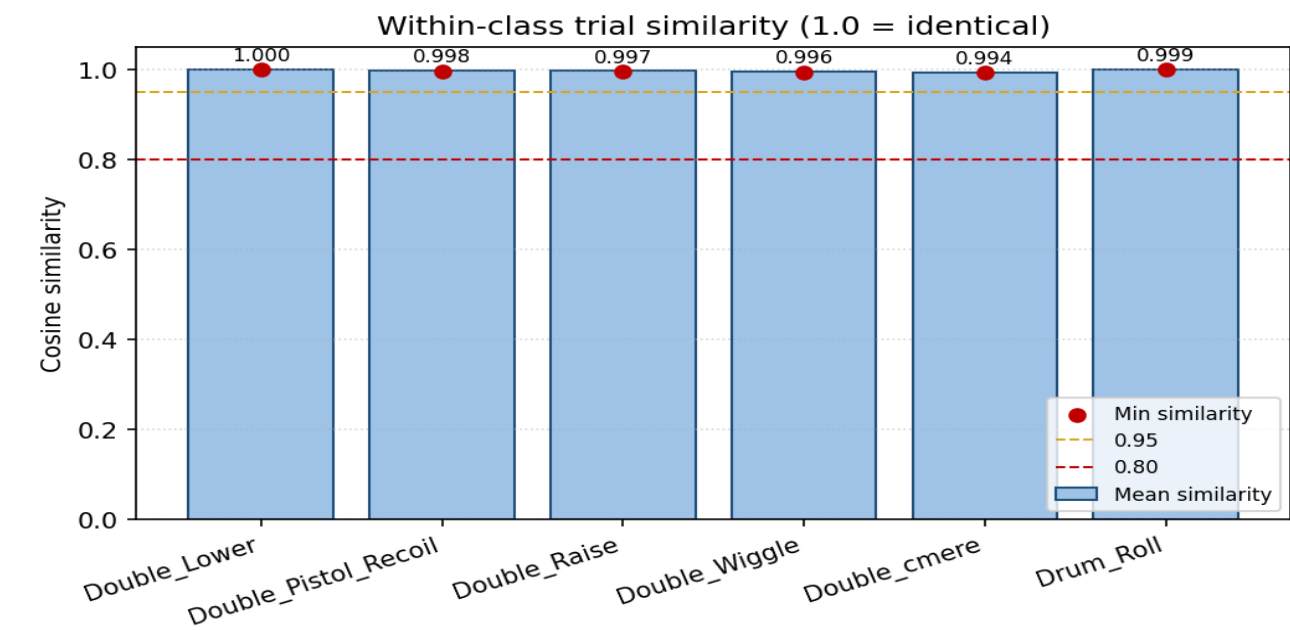


t-SNE feature space projection

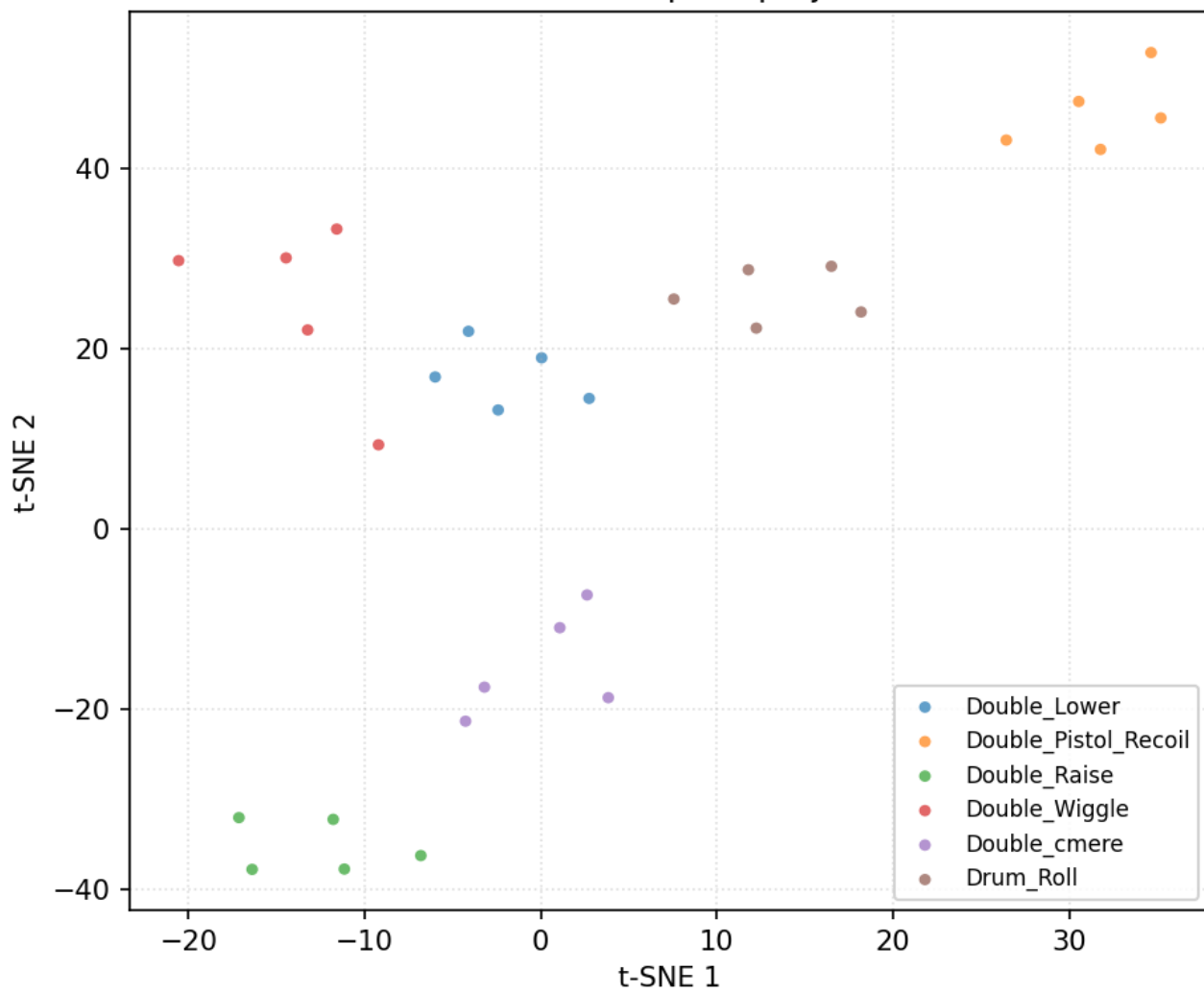


# Data diagnostics — user 16\_Bella

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

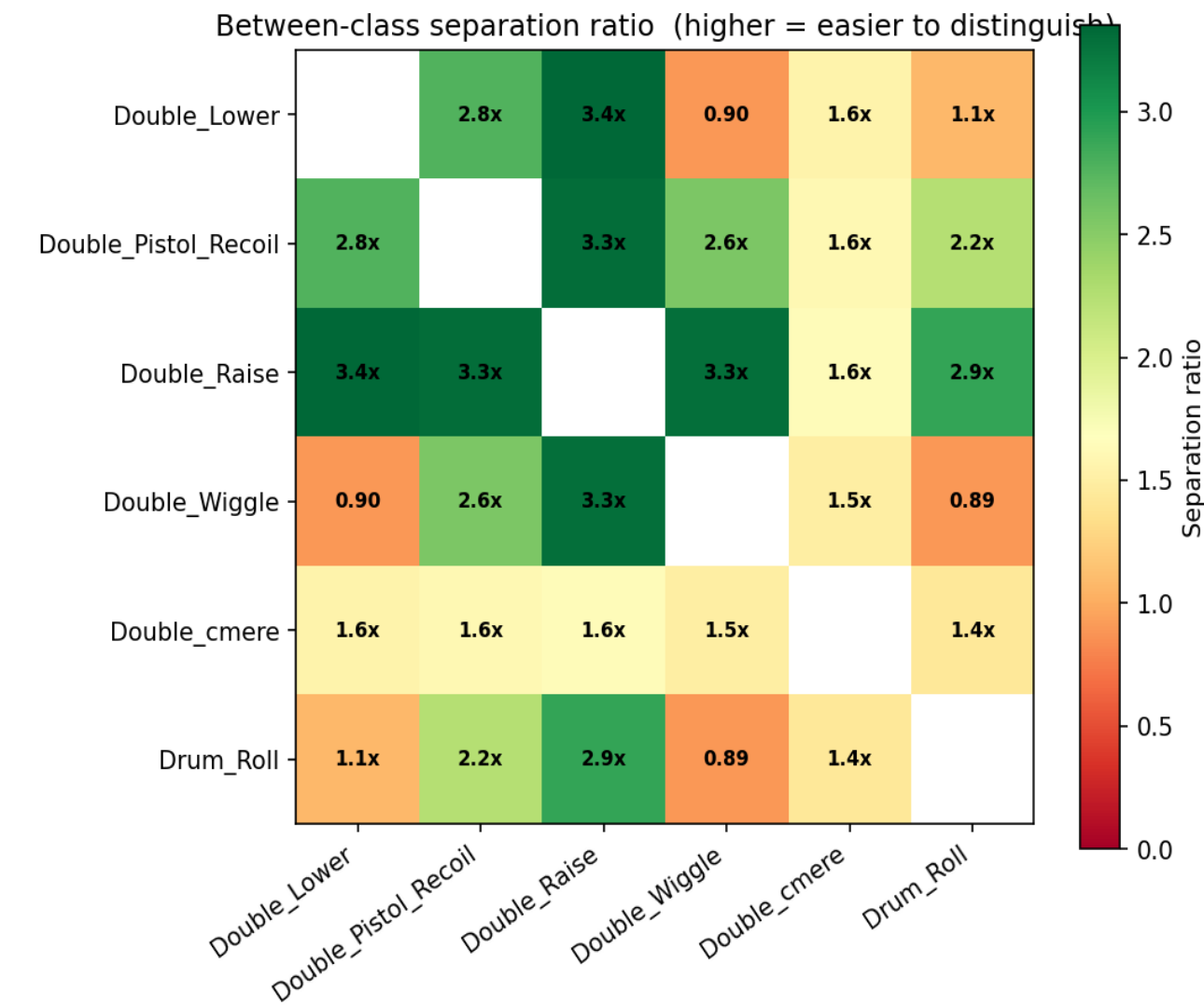
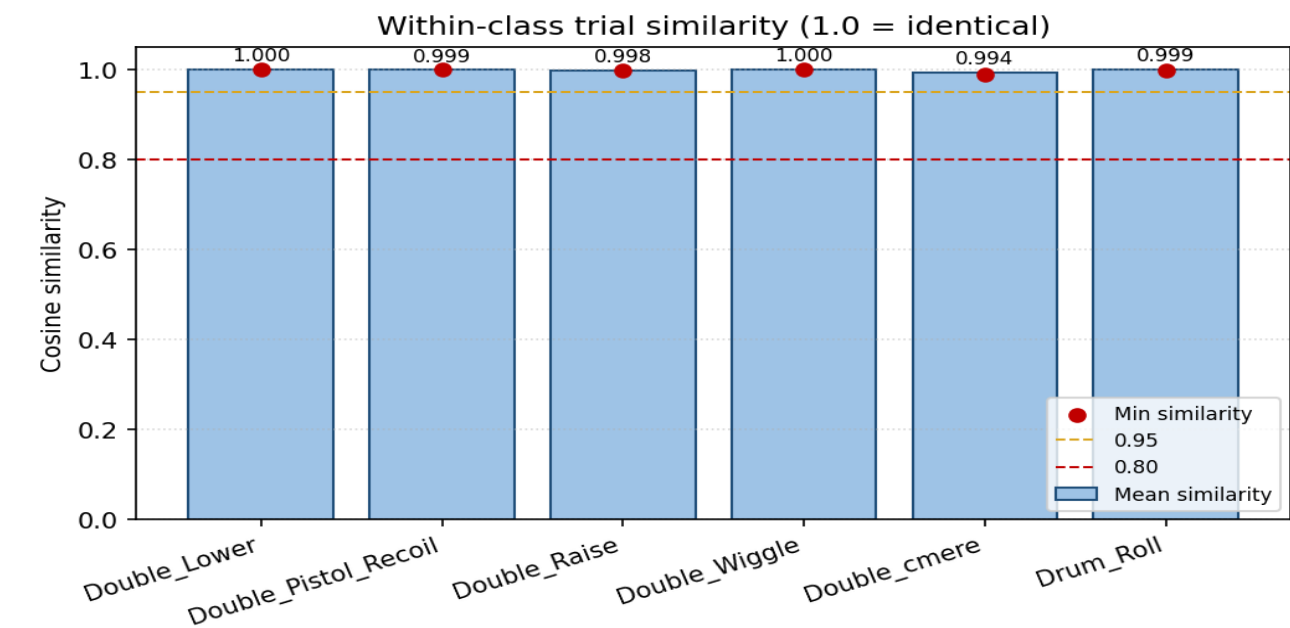


t-SNE feature space projection

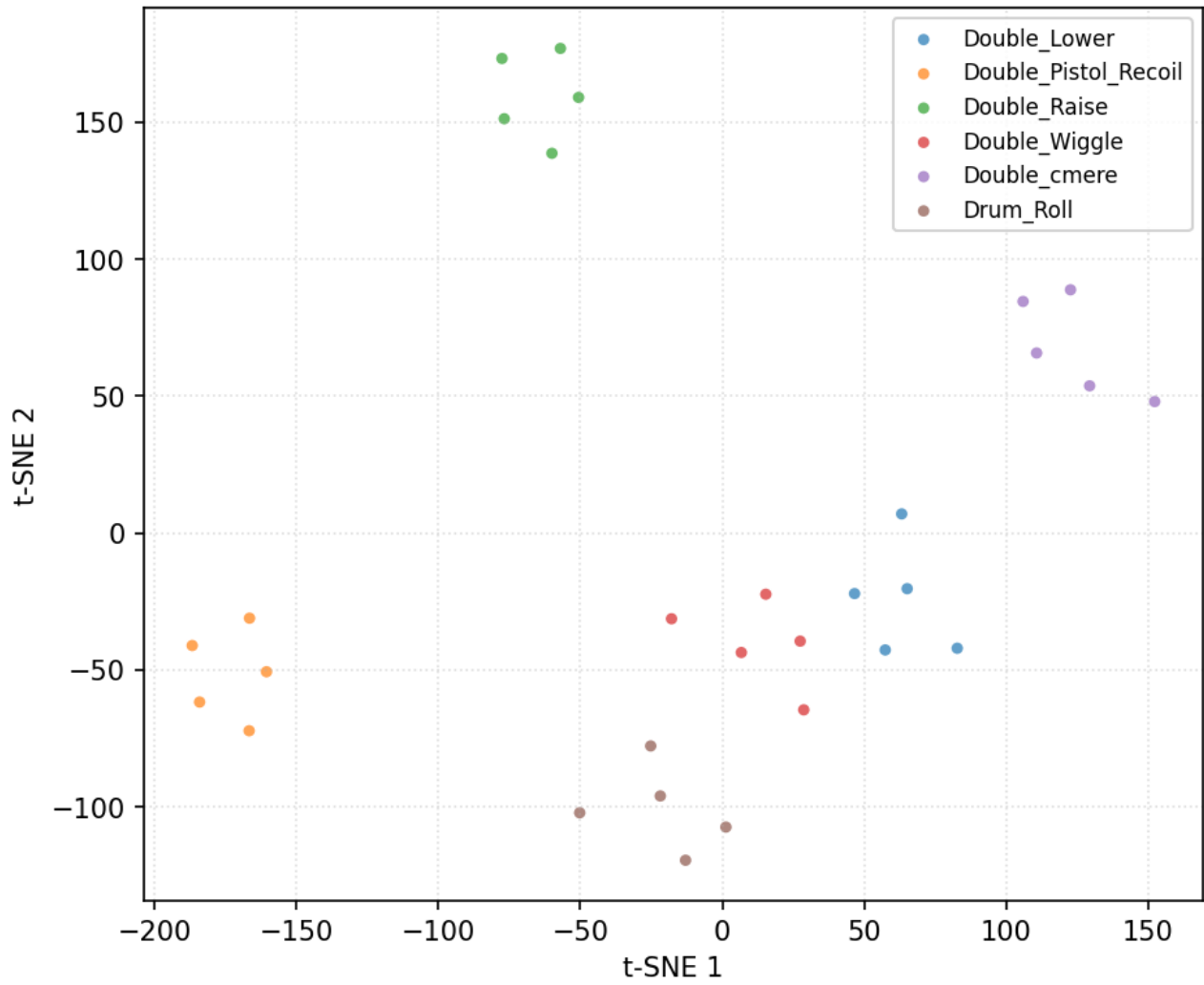


# Data diagnostics — user 17\_Raquel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).



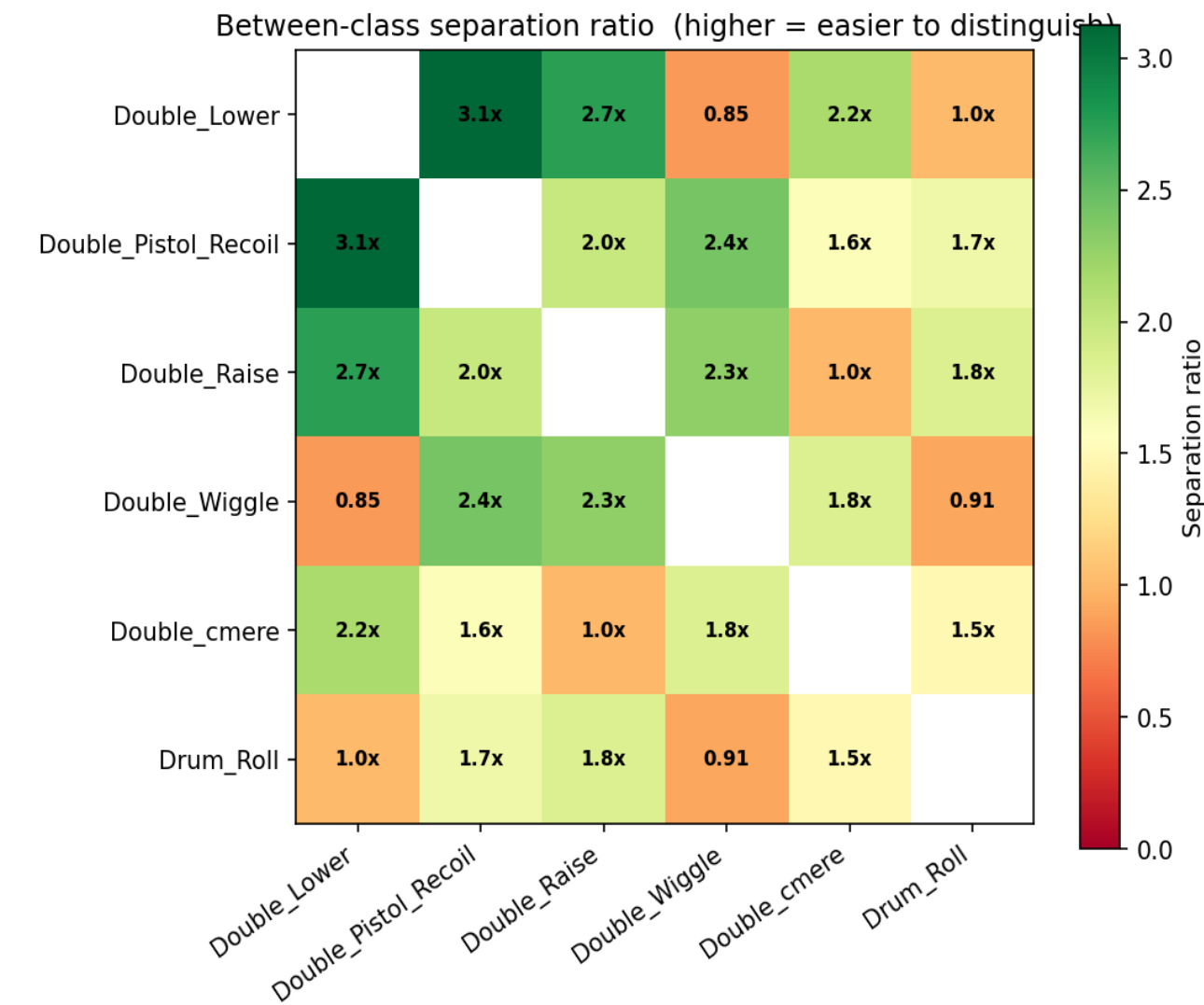
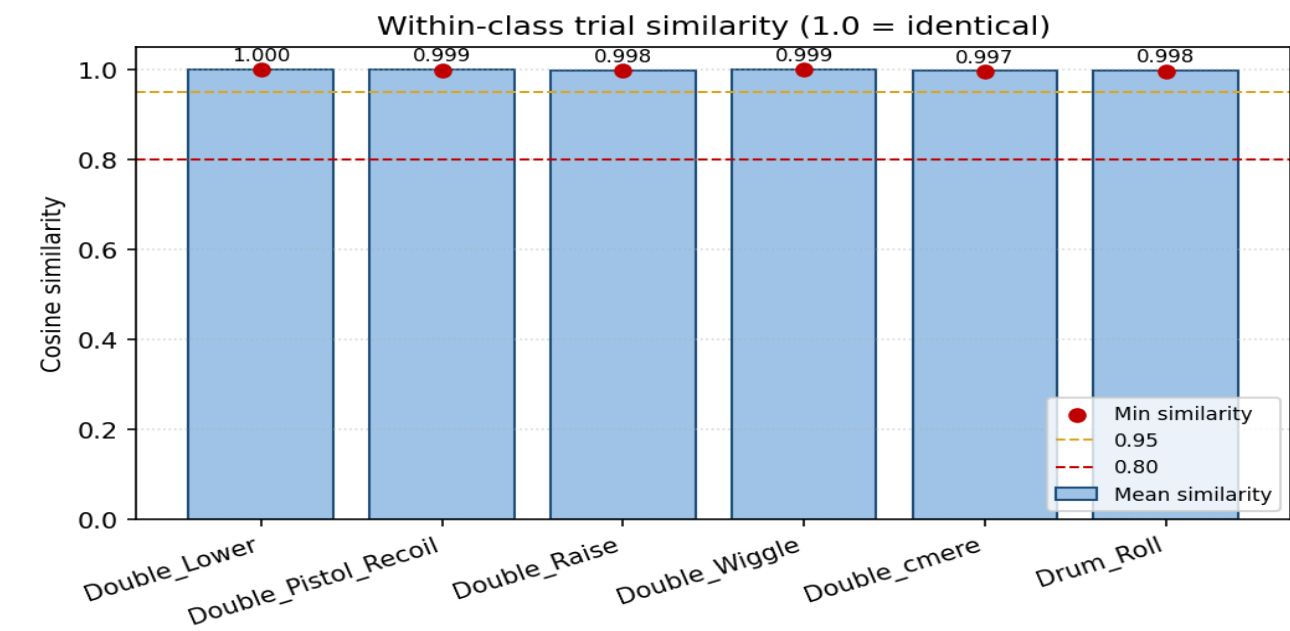
t-SNE feature space projection

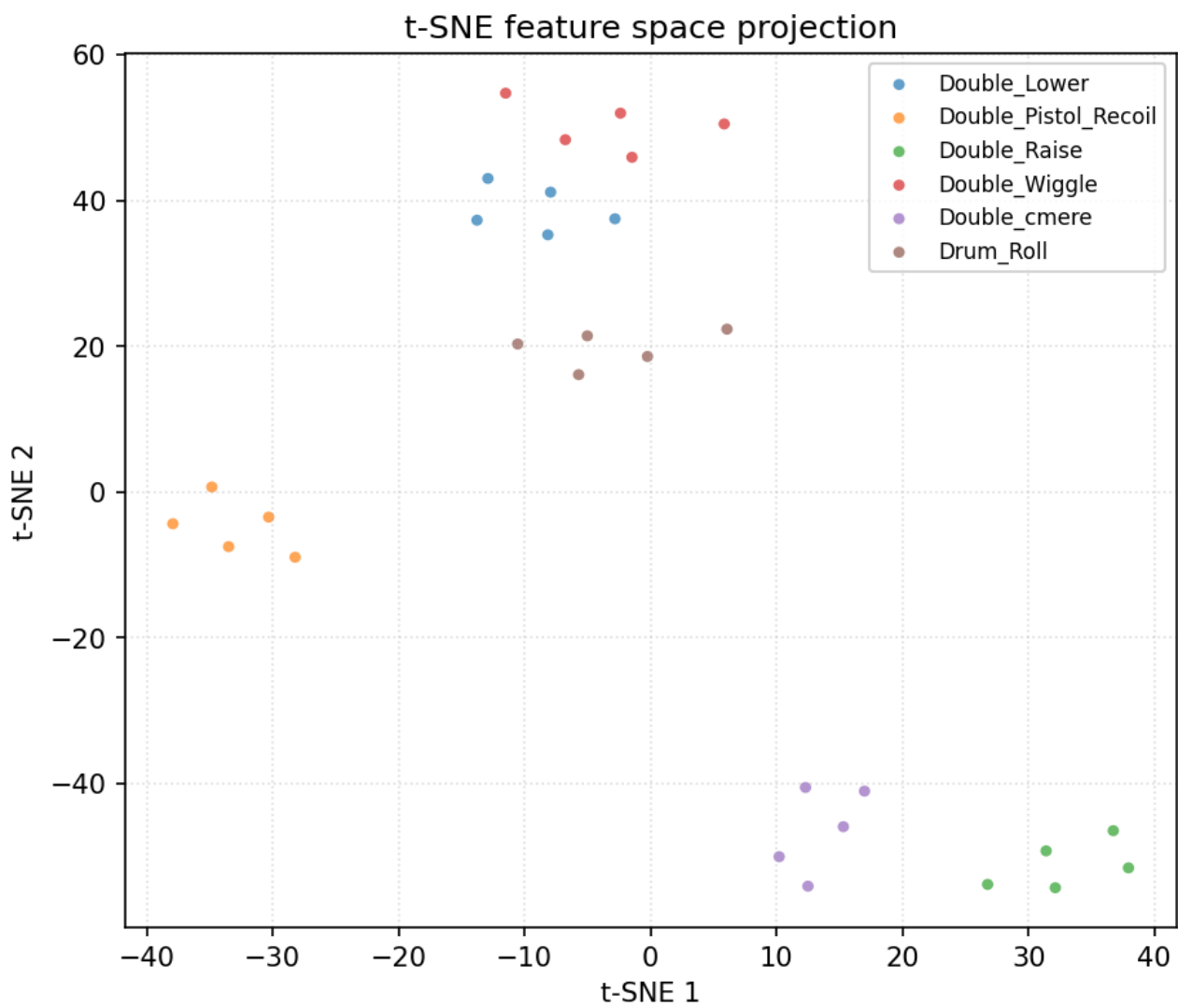




# Data diagnostics — user 18\_Bridgette

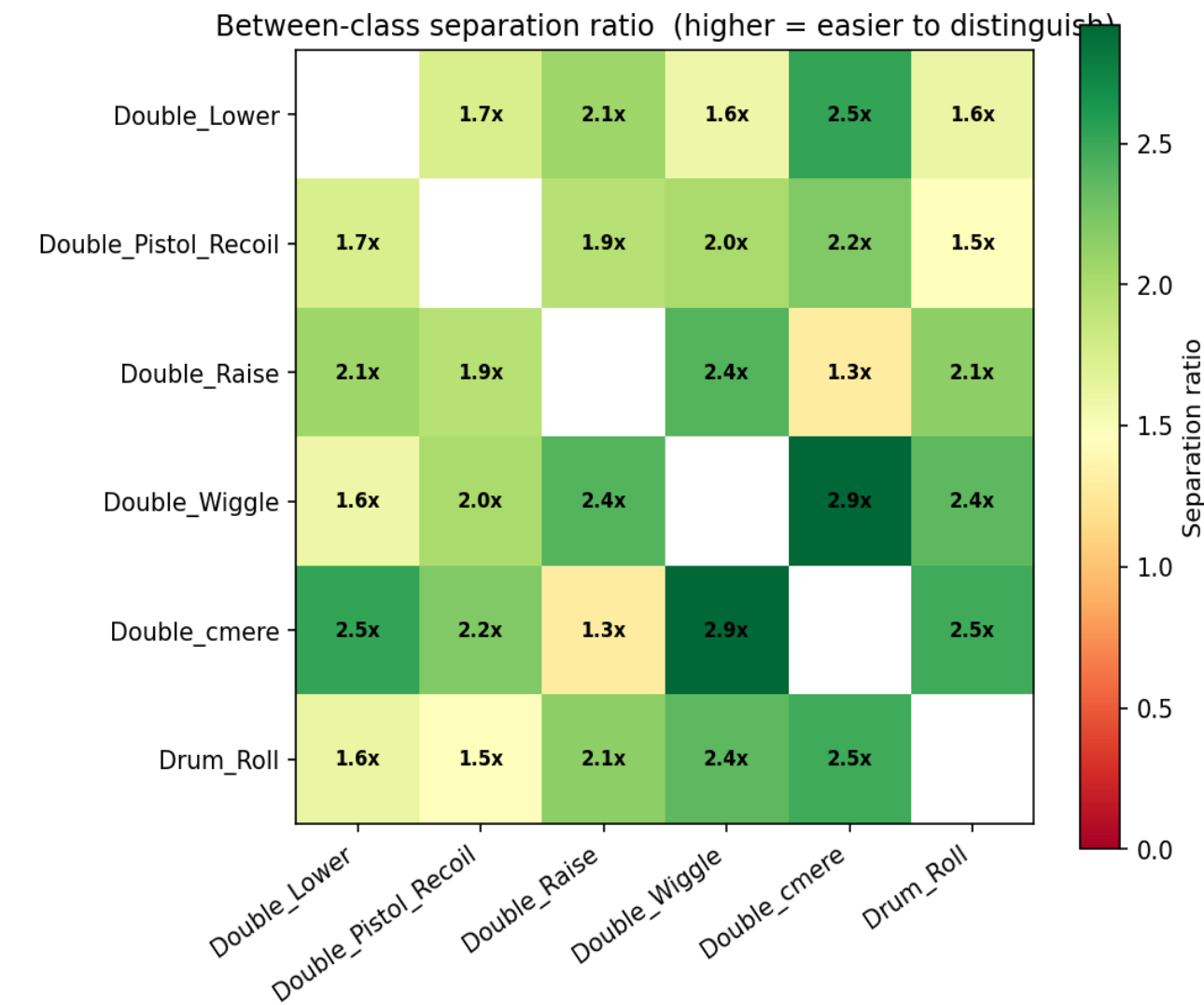
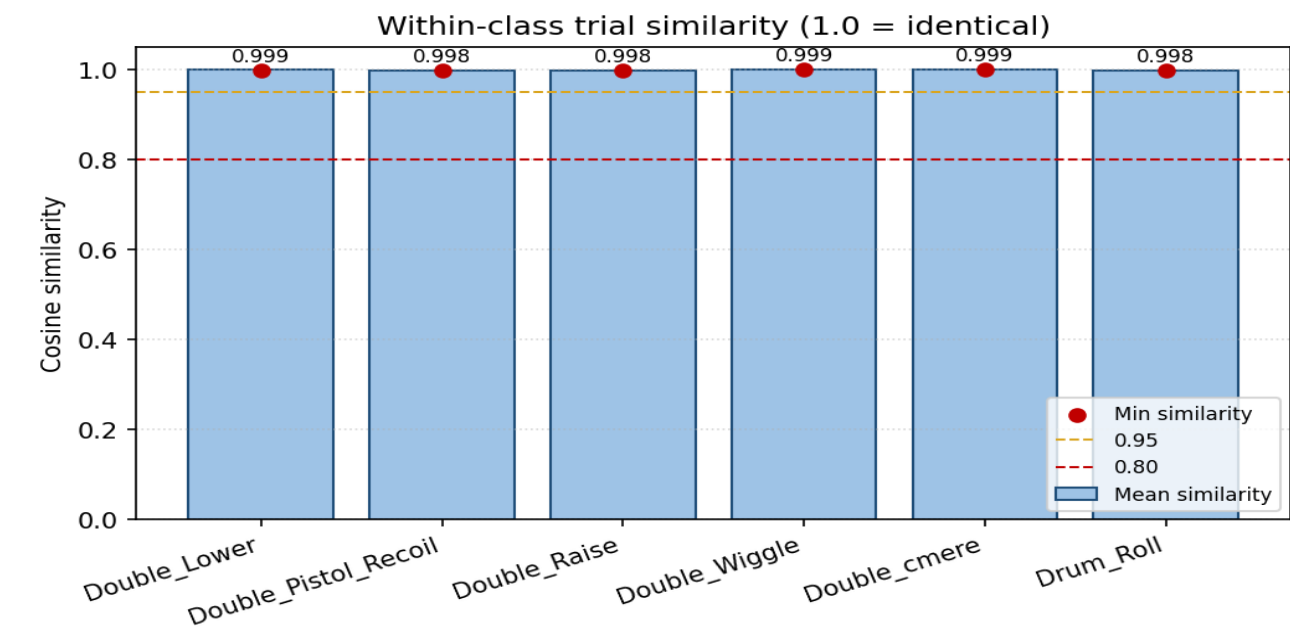
30 trials, 6 classes (filtered + resampled, before per-trial normalisation).



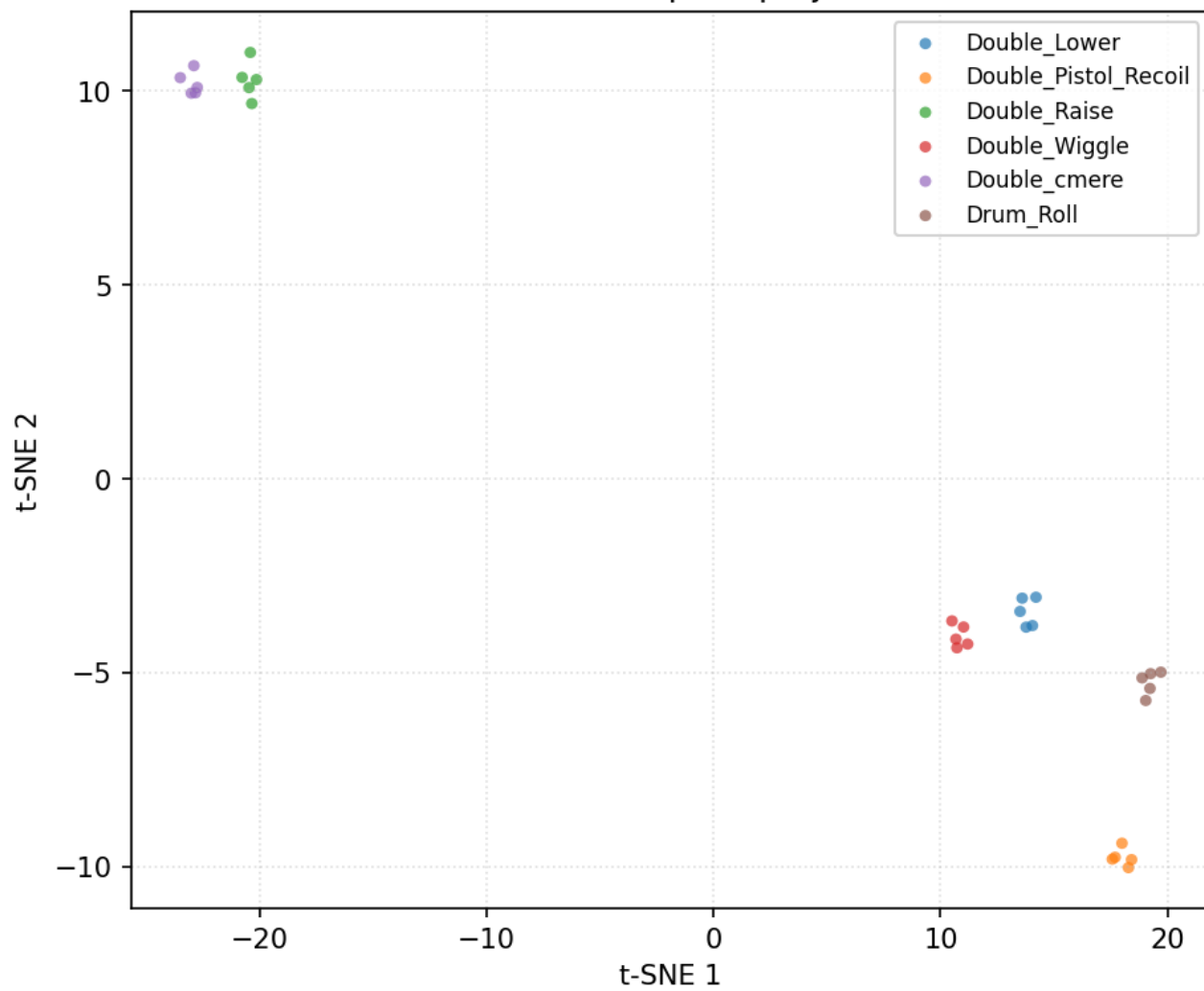


# Data diagnostics — user 1\_Alán

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

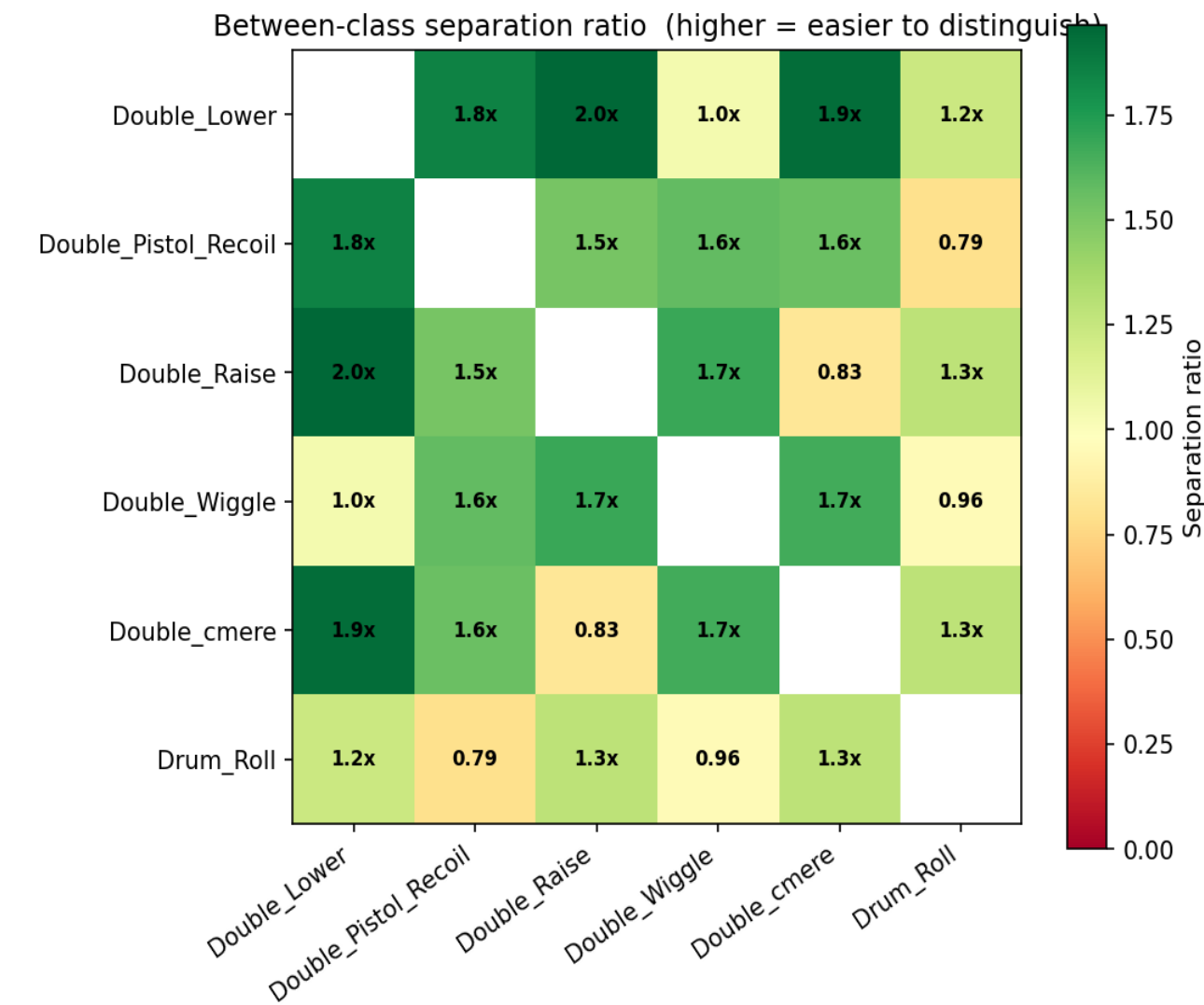
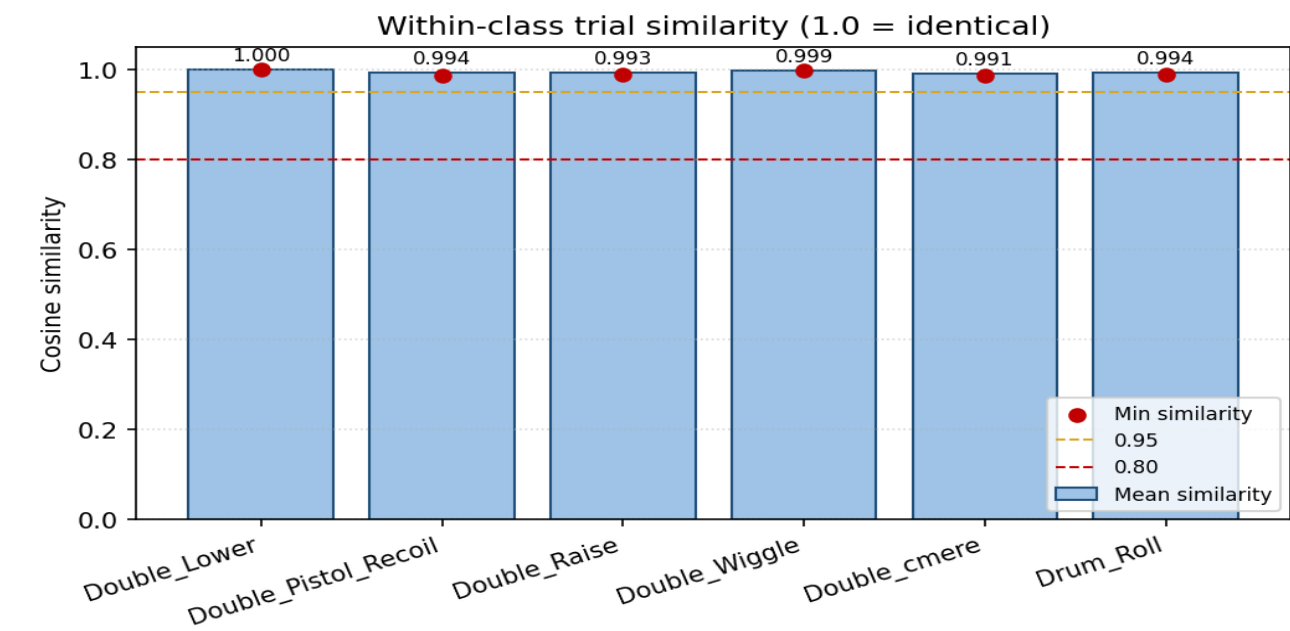


t-SNE feature space projection

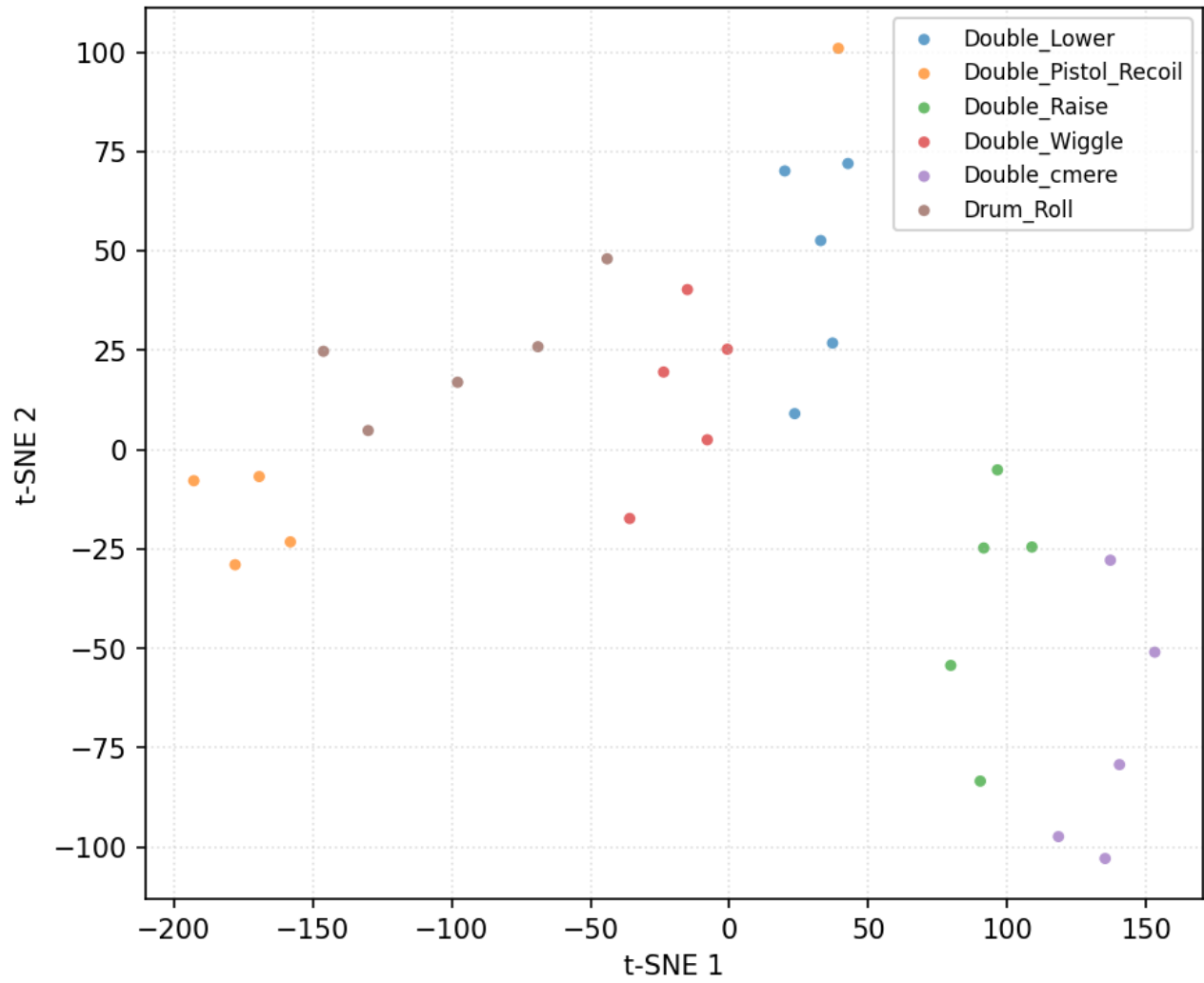


# Data diagnostics — user 2\_Alex

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

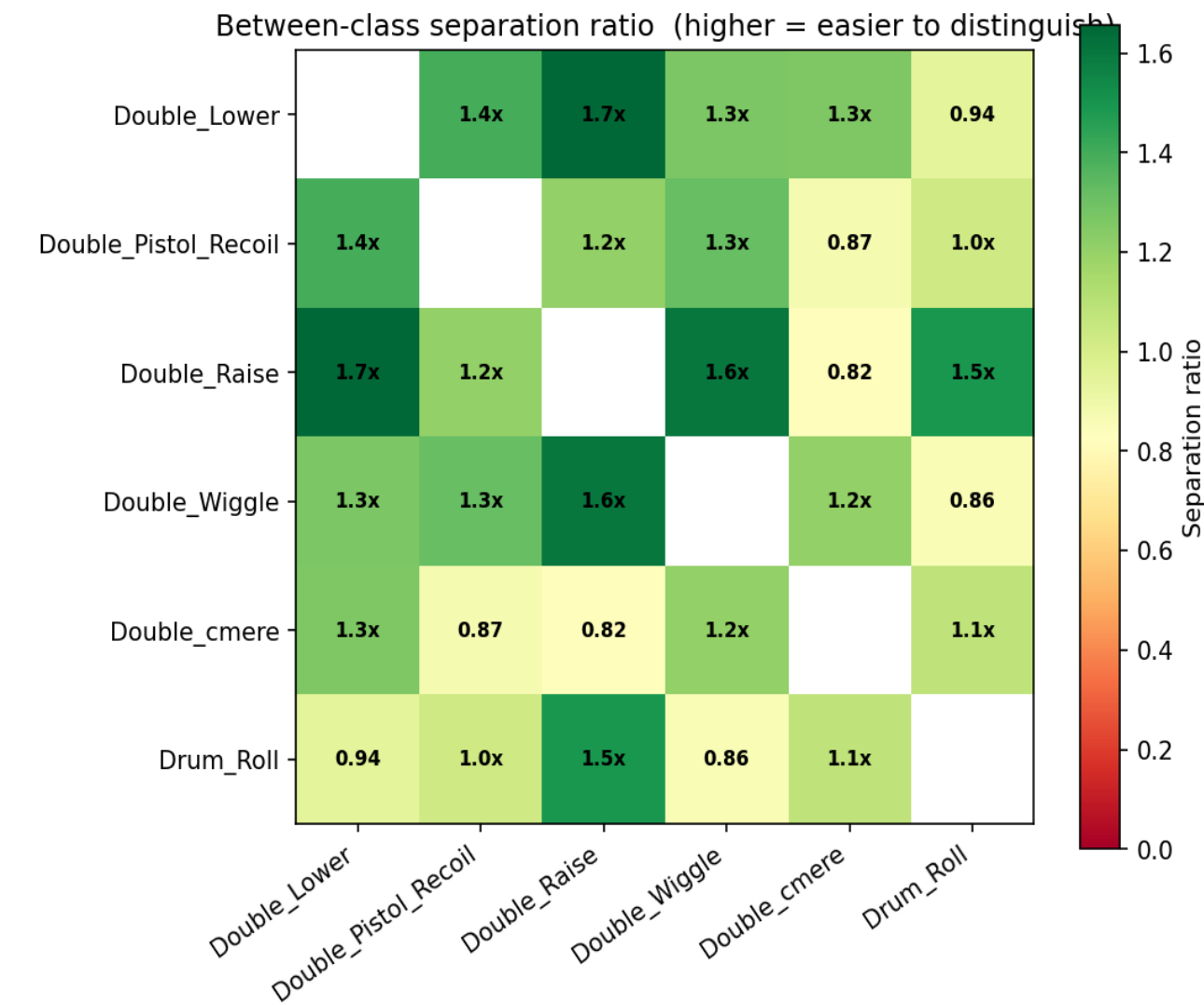
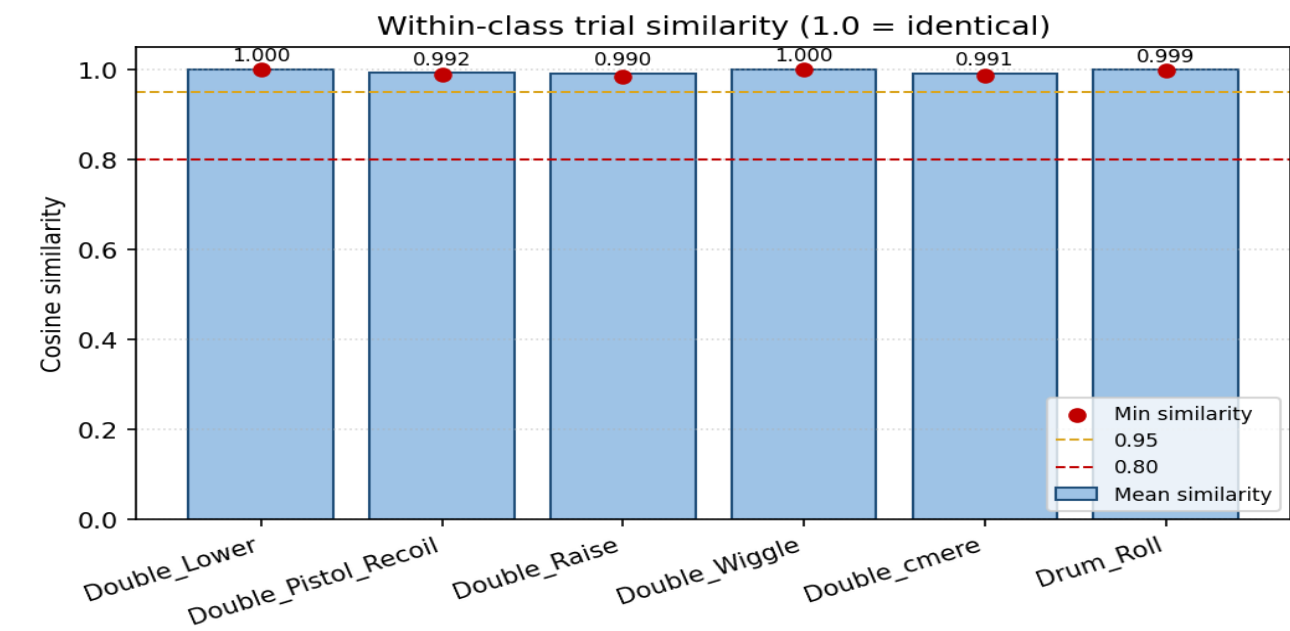


t-SNE feature space projection

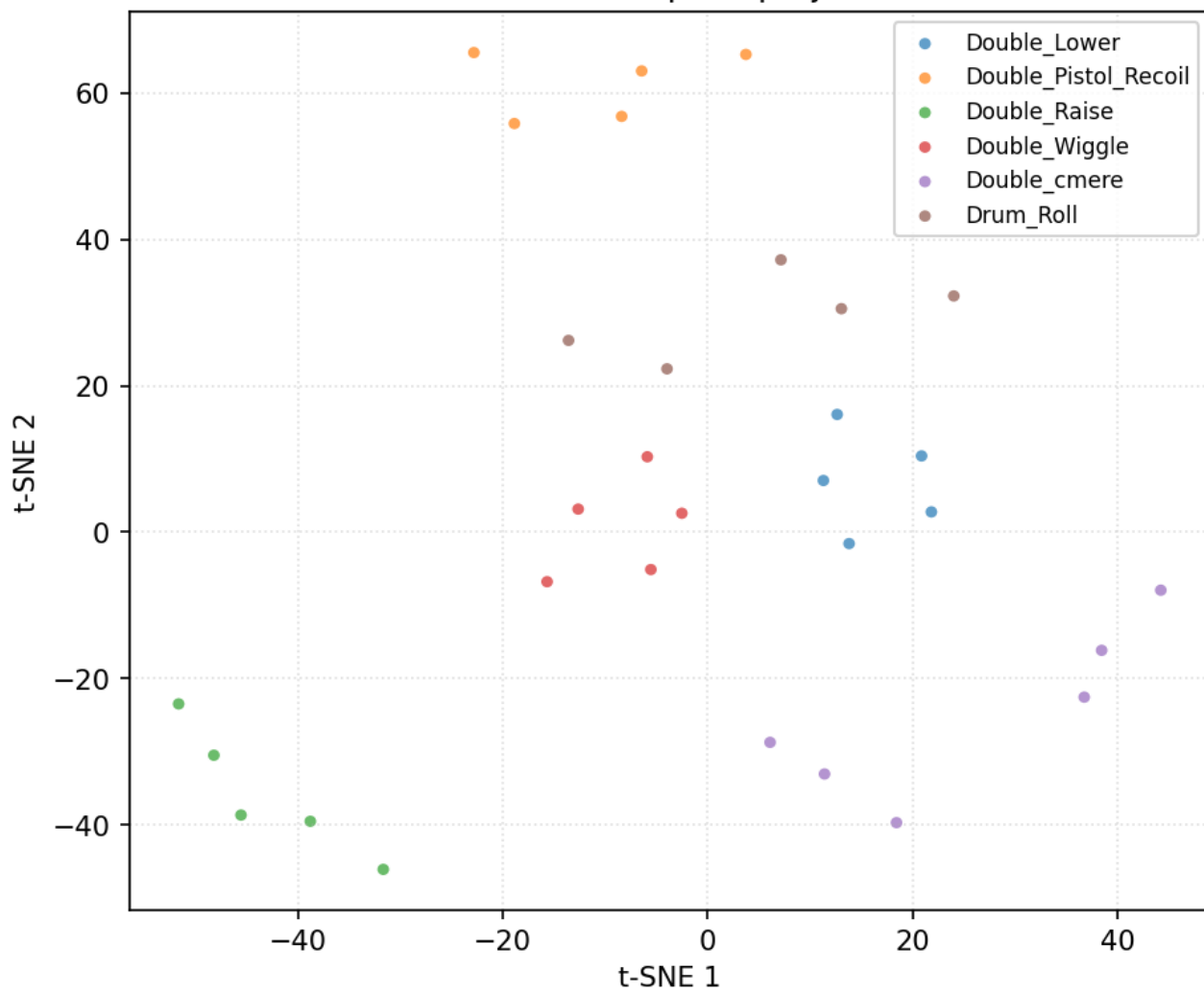


# Data diagnostics — user 3\_Anghad

31 trials, 6 classes (filtered + resampled, before per-trial normalisation).



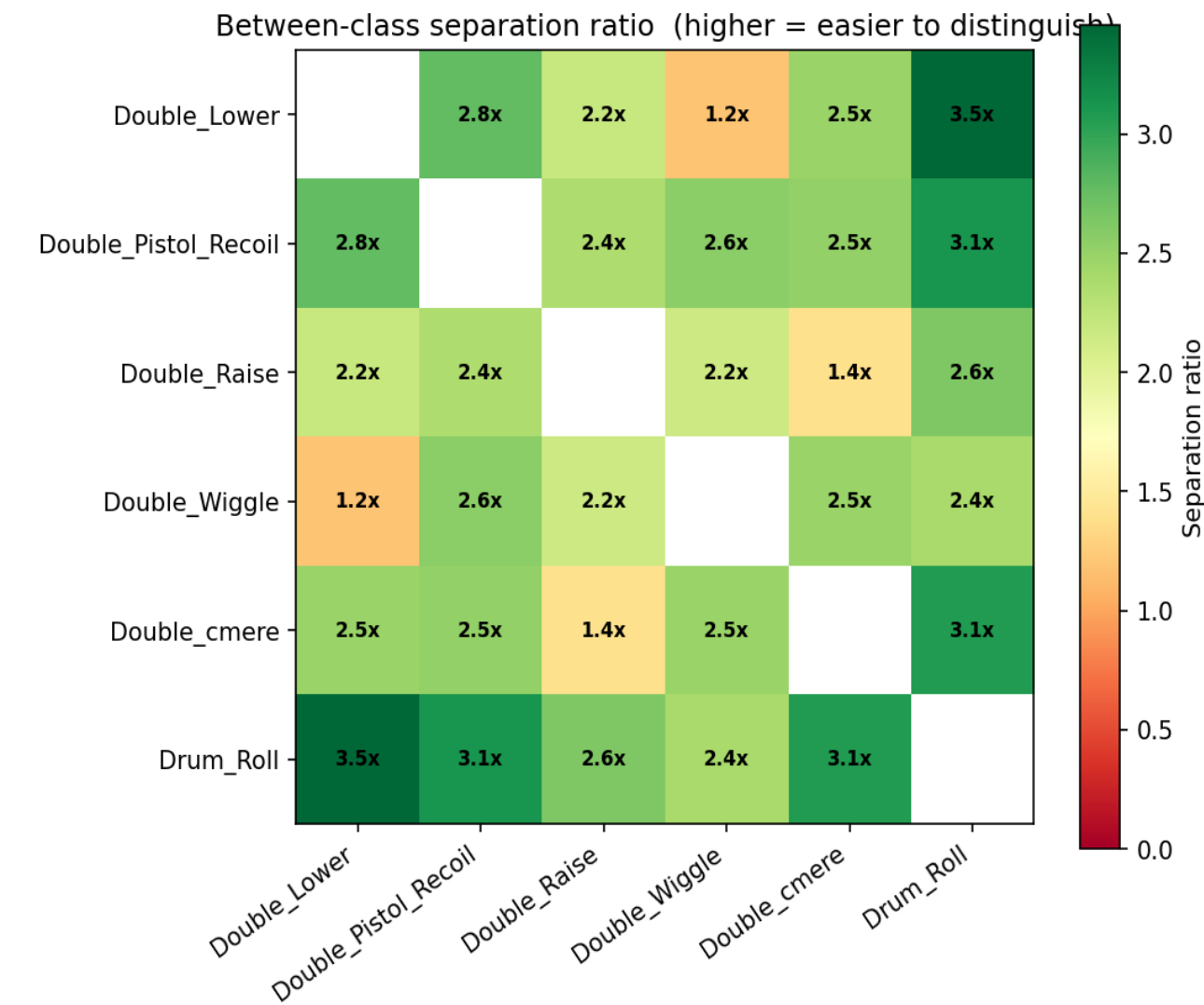
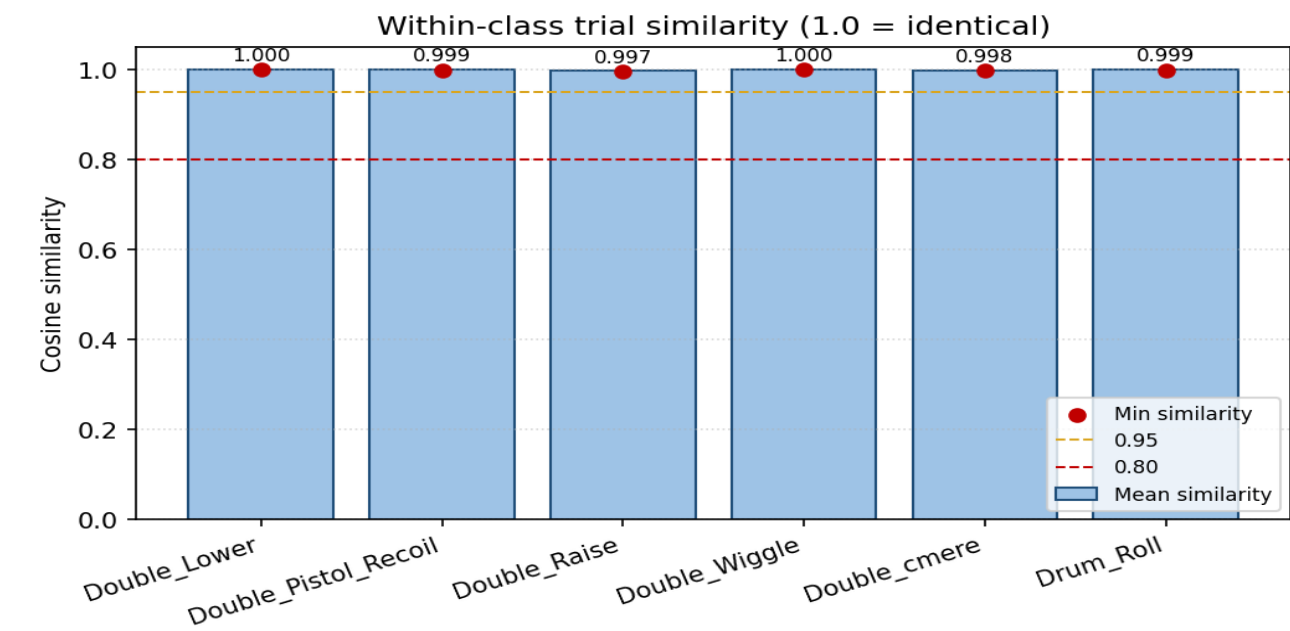
t-SNE feature space projection



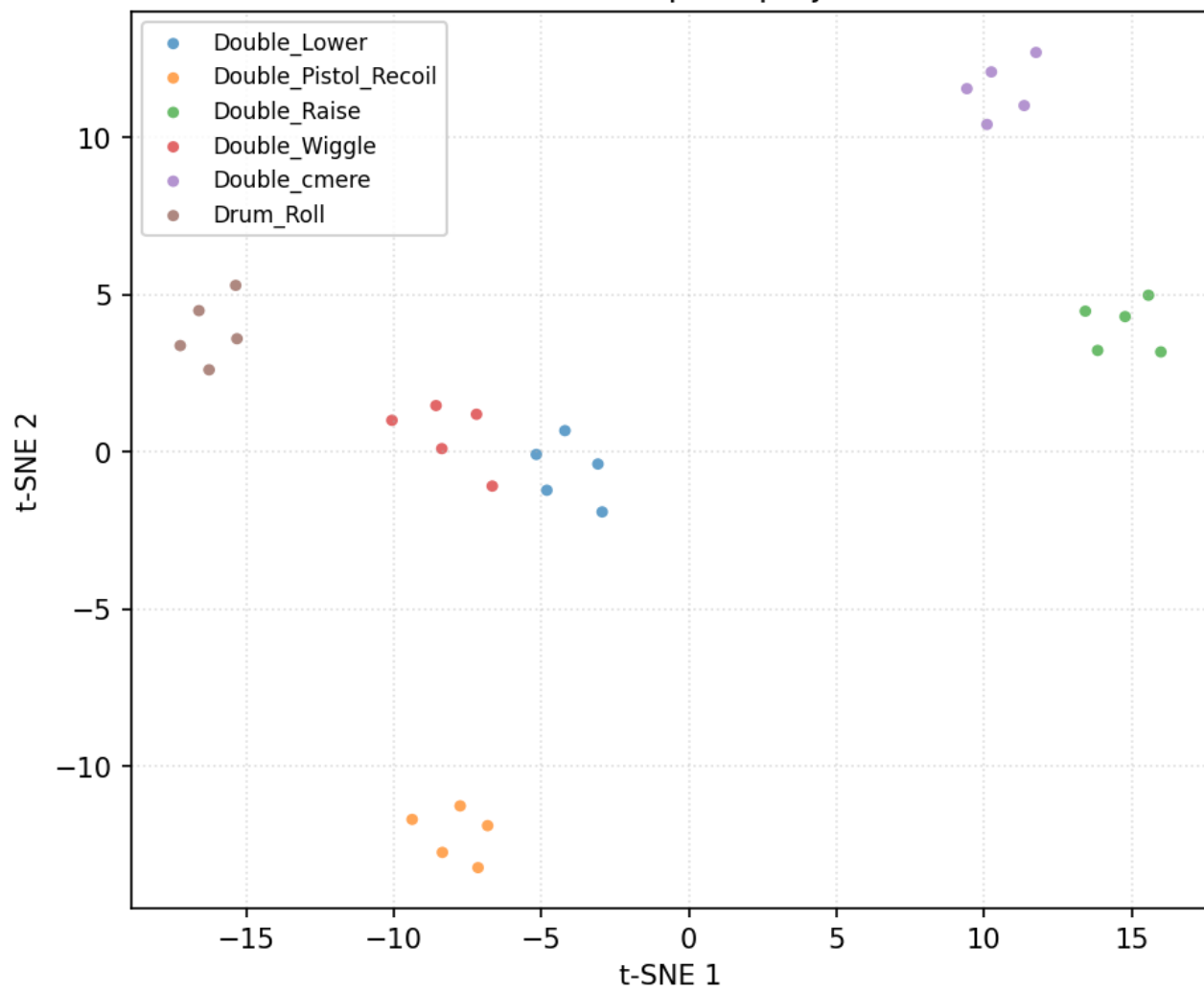


# Data diagnostics — user 4\_Daniel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

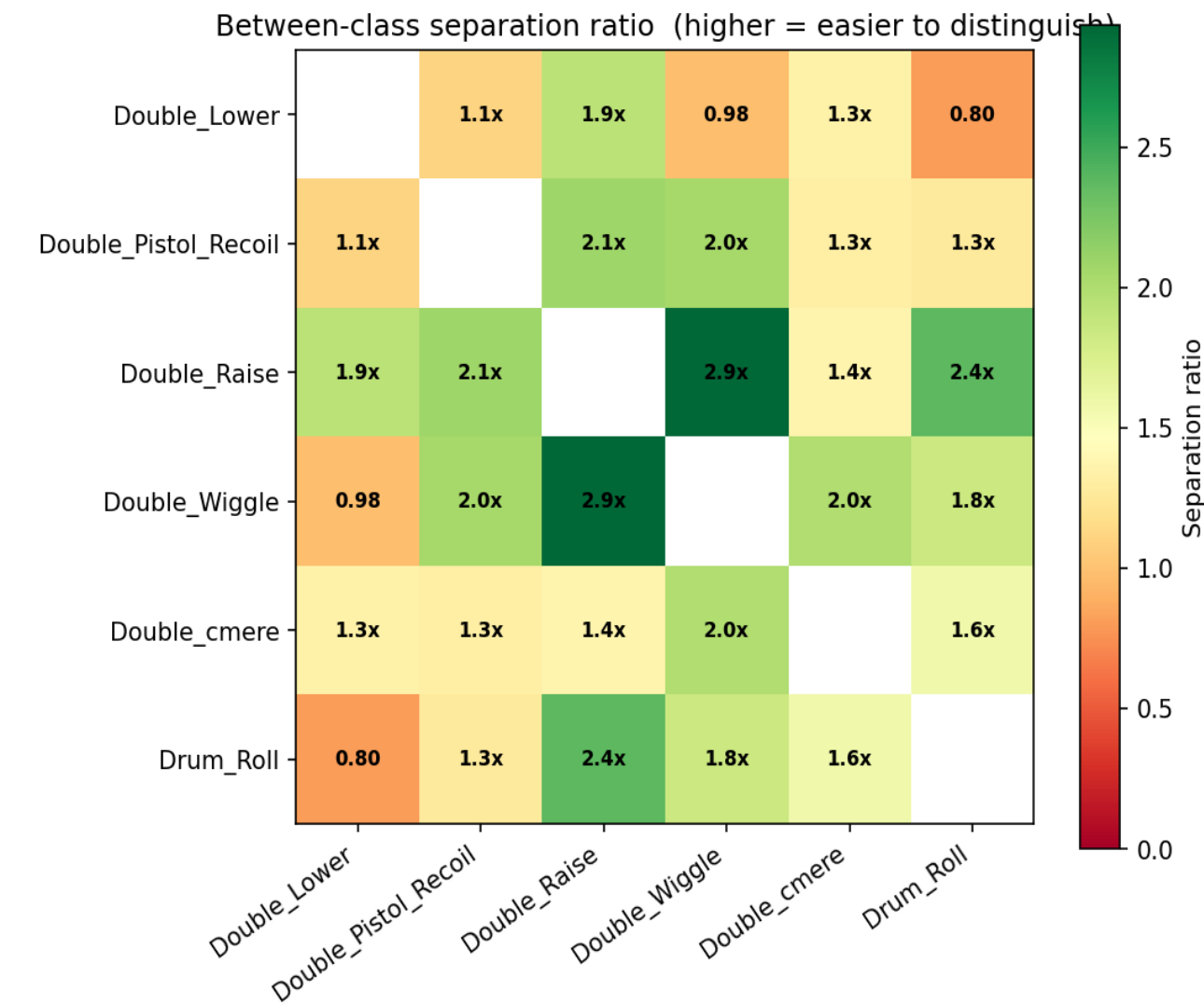
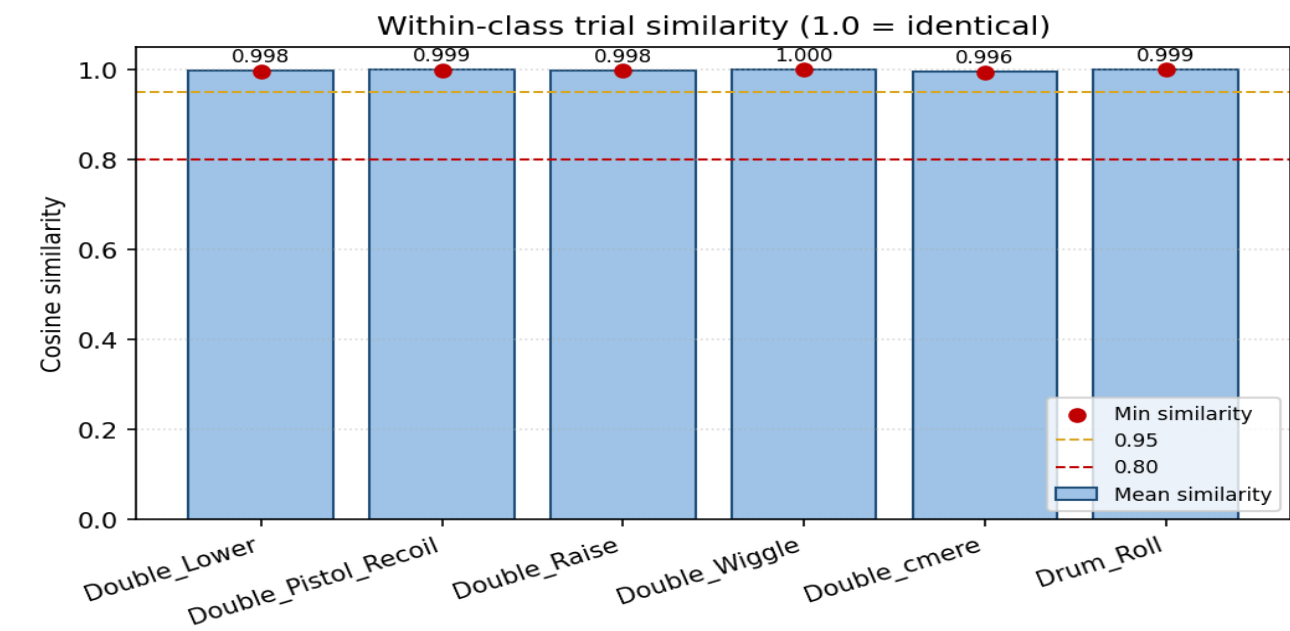


t-SNE feature space projection

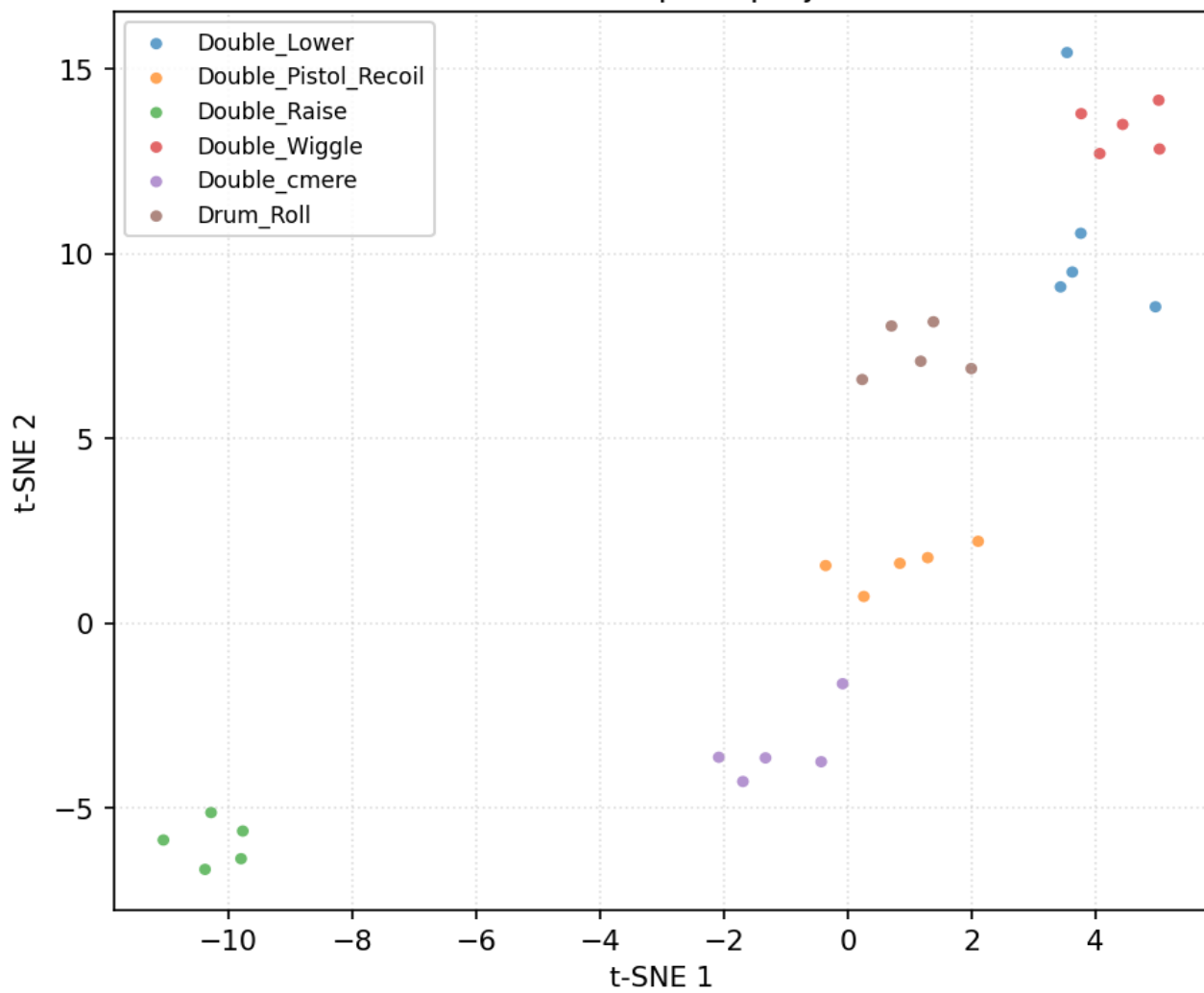


# Data diagnostics — user 6\_Harry

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

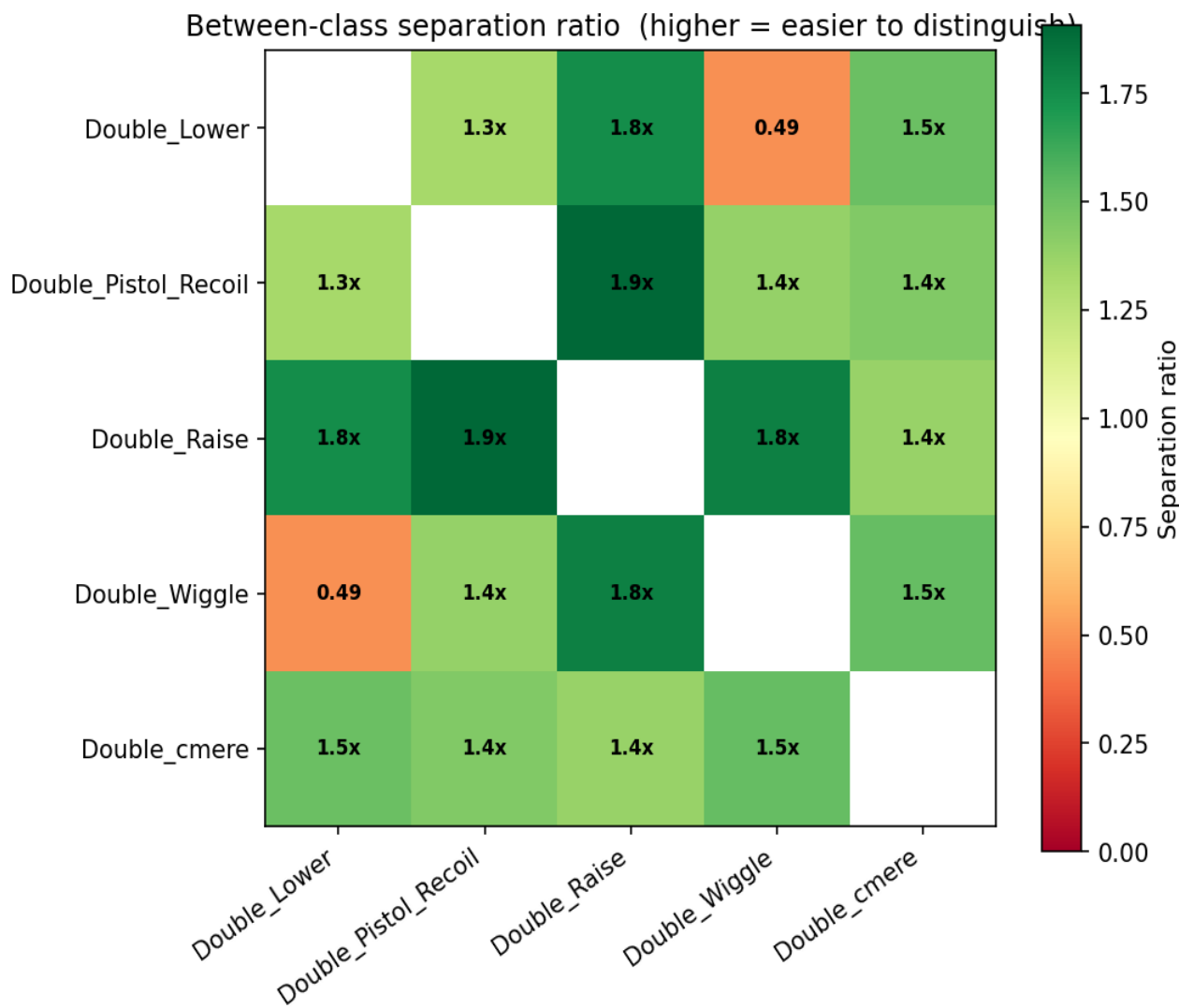
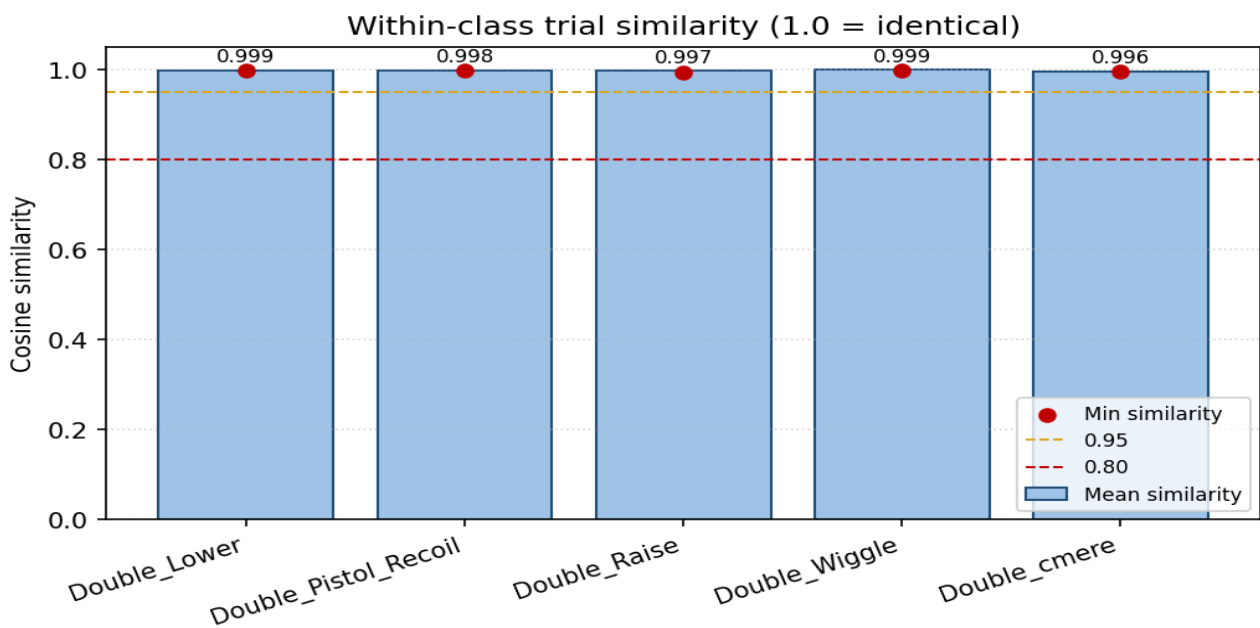


t-SNE feature space projection

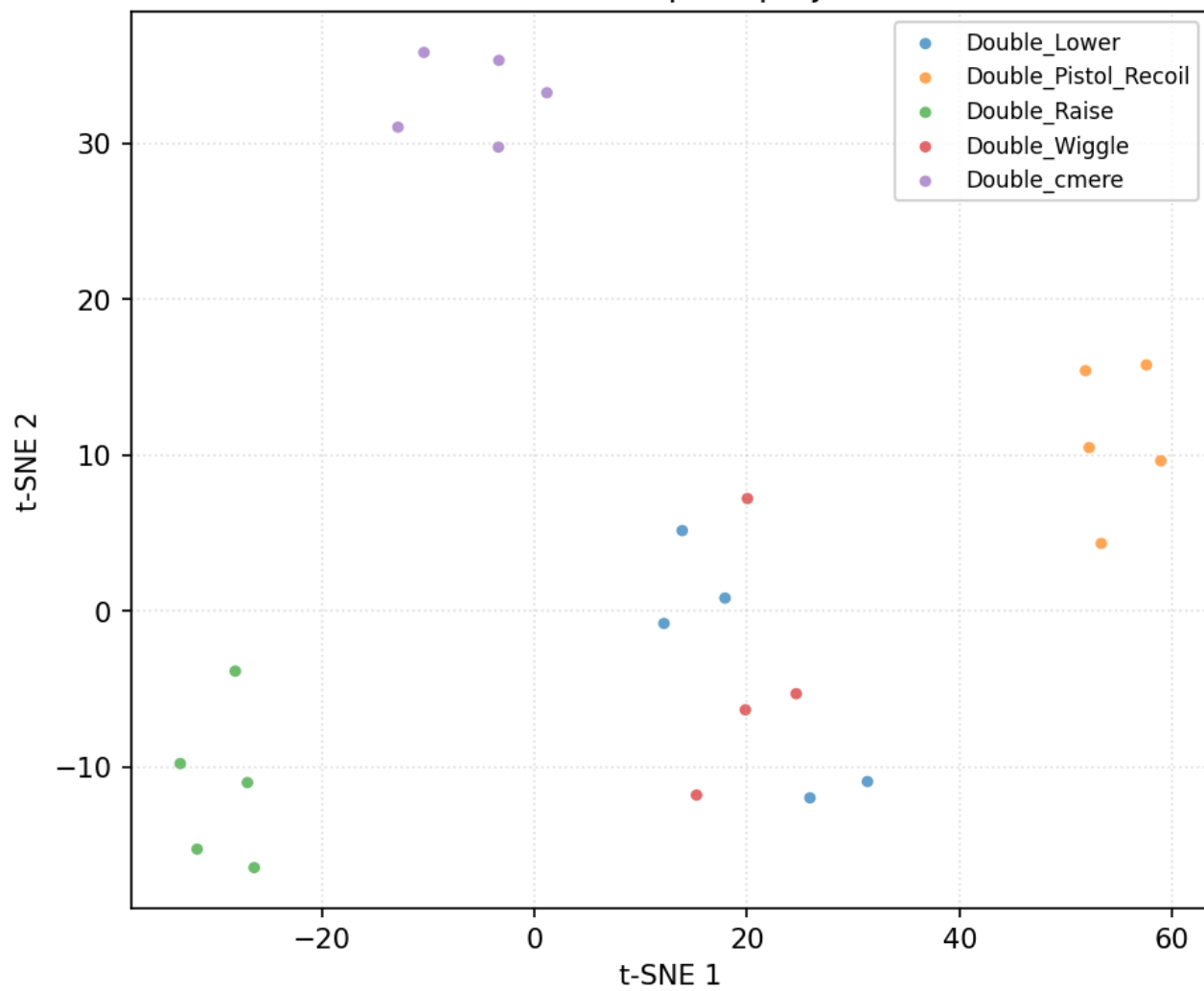


# Data diagnostics — user 8\_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



# Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

|                  |                 |
|------------------|-----------------|
| Trainable params | 113,190         |
| LOUO mean ± std  | 0.7494 ± 0.1911 |
| Folds run        | 13              |
| Inner-val mode   | user            |
| Per-trial norm   | none            |

## LOUO per-fold results

| Fold | Held-out user | Inner-val user | n_train | n_test | Accuracy |
|------|---------------|----------------|---------|--------|----------|
| 1    | 11_Mansh      | 12_Marcus      | 325     | 30     | 0.5333   |
| 2    | 12_Marcus     | 14_StephenV2   | 325     | 30     | 0.7333   |
| 3    | 14_StephenV2  | 15_Tash        | 325     | 30     | 1.0000   |
| 4    | 15_Tash       | 16_Bella       | 325     | 30     | 0.7667   |
| 5    | 16_Bella      | 17_Raquel      | 325     | 30     | 0.6000   |
| 6    | 17_Raquel     | 18_Bridgette   | 325     | 30     | 0.6667   |
| 7    | 18_Bridgette  | 1_Alan         | 325     | 30     | 0.5000   |
| 8    | 1_Alan        | 2_Alex         | 325     | 30     | 0.6000   |
| 9    | 2_Alex        | 3_Anghad       | 324     | 30     | 0.9667   |
| 10   | 3_Anghad      | 4_Daniel       | 324     | 31     | 1.0000   |
| 11   | 4_Daniel      | 6_Harry        | 325     | 30     | 1.0000   |
| 12   | 6_Harry       | 8_Jestin       | 331     | 30     | 0.5000   |
| 13   | 8_Jestin      | 11_Mansh       | 331     | 24     | 0.8750   |

## Per-class metrics (aggregated across folds)

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.713     | 0.877  | 0.786 | 65      |
| Double_Pistol_Recoil | 0.922     | 0.908  | 0.915 | 65      |
| Double_Raise         | 0.757     | 0.815  | 0.785 | 65      |
| Double_Wiggle        | 0.629     | 0.609  | 0.619 | 64      |
| Double_cmere         | 0.803     | 0.742  | 0.772 | 66      |
| Drum_Roll            | 0.646     | 0.517  | 0.574 | 60      |
| macro avg            | 0.745     | 0.745  | 0.742 | 385     |
| weighted avg         | 0.747     | 0.748  | 0.744 | 385     |

## Confusion matrix (aggregated across folds)

